PRODUCT REQUIREMENTS & SOLUTION SPECIFICATION

AI CHANGE IMPACT ANALYZER

AI-assisted decision support for understanding the ripple effects of organizational change.

Product Vision · Business Analysis · Requirements · AI Architecture · Evaluation · Roadmap



Version 1.0 | September2026

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Product Vision · Business Analysis · Requirements · AI Architecture · Evaluation · Roadmap

PRODUCT DOCUMENTATION

DOCUMENT CONTROL

Field	Details
Product	AI Change Impact Analyzer
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Purpose of this Document

This document defines the product vision, problem space, target users, functional requirements, AI capabilities, user experience, evaluation approach, and MVP scope for the AI Change Impact Analyzer.

Executive Summary

Organizations frequently introduce changes to processes, technologies, policies, operating models, and workflows. While the requested change may appear limited in scope, its effects can propagate across teams, business processes, systems, data, documentation, dependencies, and operational risks.

The AI Change Impact Analyzer is an AI-assisted decision-support product designed to help Business Analysts, Project Managers, and Transformation teams identify and assess these potential impacts before implementation begins.

The product accepts a proposed organizational change and relevant contextual information, analyzes the change across predefined impact dimensions, and produces a structured impact assessment covering affected areas, severity, confidence, rationale, risks, dependencies, and recommended actions.

The system is designed to augment human decision-making rather than replace it. AI-generated findings are presented with contextual reasoning and confidence indicators so that users can review, validate, and refine the analysis before making implementation decisions.

01	CHANGE	⇢	Proposed organizational change
02	IMPACT	⇢	Processes • People • Systems • Data
03	RISK	⇢	Dependencies • Uncertainty • Consequences
04	ACTION	⇢	Recommendations • Validation • Next steps

CHANGE → IMPACT → RISK → ACTION

Product Principle

AI should expand the analyst's field of view—not replace the analyst's judgment.

Problem Statement

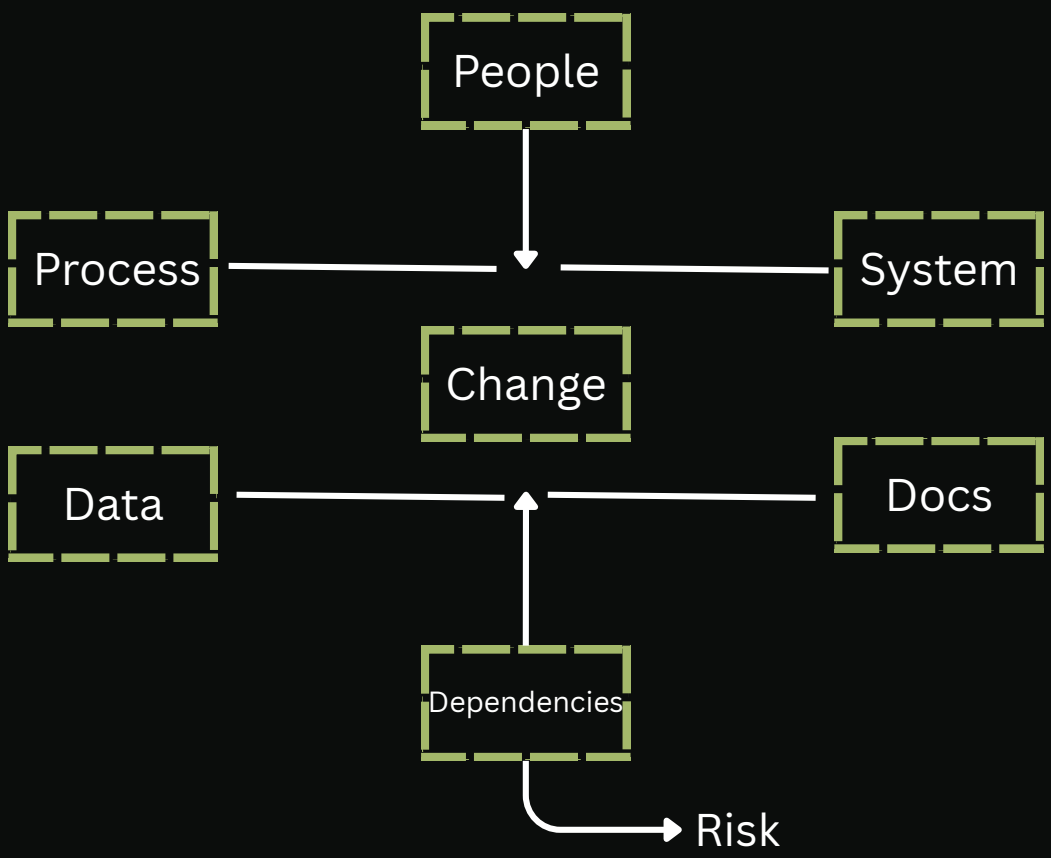
*Understanding the **ripple effects** of change*

Organizational changes rarely remain isolated to the area in which they are introduced.

A change to a business process, technology, policy, workflow, or operating model can create downstream effects across people, processes, systems, data, documentation, dependencies, and operational risks.

However, early-stage change impact analysis is often dependent on manual investigation, stakeholder knowledge, existing documentation, and the individual experience of the analyst. This can make the process time-consuming, inconsistent, and vulnerable to overlooked dependencies.

The opportunity is to use AI as an assistive analytical layer that helps surface potential impacts systematically while keeping human experts responsible for validation and decision-making.



Current-State Analysis

Why existing analysis can be difficult to scale

Current-State Workflow

Change Request → Initial review → Document review → Stakeholder Discussions → Impact Identification → Risk & Dependency analysis → Impact Report

The challenge: *Much of this analysis relies on fragmented organizational knowledge and manual investigation.*

Key Pain Points

01 — Fragmented Knowledge

Relevant information may be distributed across process documents, SOPs, system documentation, stakeholder knowledge, and previous project artifacts.

02 — Hidden Dependencies

Indirect relationships between processes, teams, systems, and data may not be immediately visible during initial analysis.

03 — Analyst Dependency

The quality and breadth of impact analysis can vary depending on the analyst's experience, domain knowledge, and access to stakeholders.

04 — Time-Intensive Investigation

Identifying and validating potential impacts can require multiple stakeholder conversations and manual review of organizational documentation.

Product Opportunity

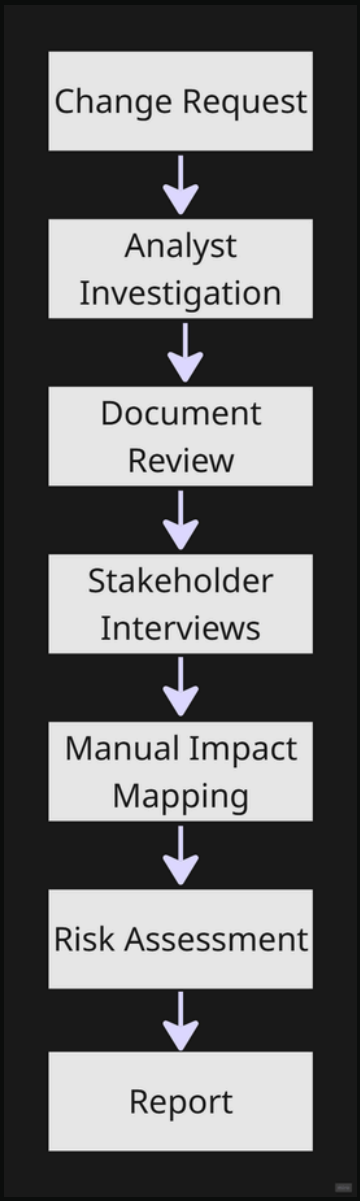
How might we help analysts identify potential change impacts earlier, more systematically, and with greater transparency—without removing human judgment from the decision-making process?

Product Opportunity

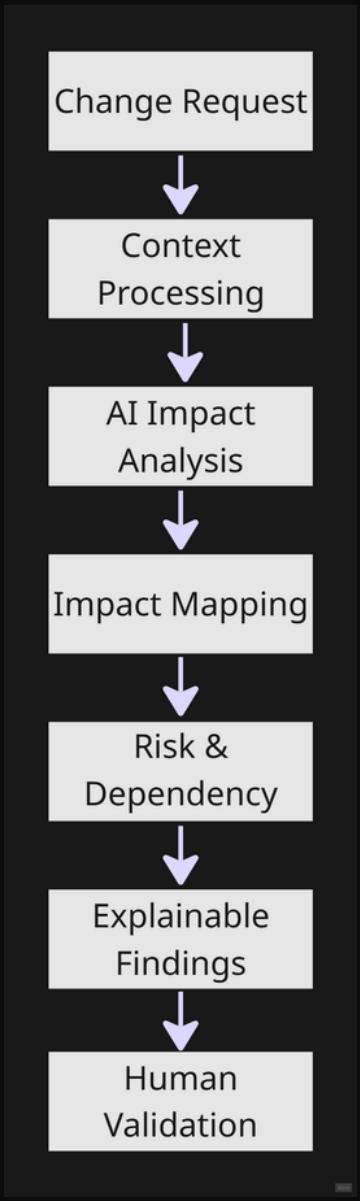
From manual investigation to AI-assisted analysis

Advances in generative AI and structured information extraction create an opportunity to augment traditional change impact analysis with an intelligent analytical layer.

Manual Change Analysis



AI-Assisted Change Analysis



Key product opportunity

The product does not aim to automate the analyst. It aims to expand the analyst's field of view.

Product Vision & Principles

Understanding change before implementing it.

Mission

Transform fragmented and largely manual change impact analysis into a structured, explainable, and AI-assisted decision-support experience.

Help analysts identify potential impacts, risks, and dependencies earlier while keeping human experts responsible for validation and decision-making.

01 — Assist, Don't Replace

AI augments the analyst's expertise by surfacing potential impacts and risks. It does not autonomously approve, reject, or implement organizational changes.

02 — Explain the "Why"

Significant findings should include understandable rationale, allowing users to assess why a process, system, team, or dependency was identified.

03 — Surface Uncertainty

The product distinguishes potential impacts from validated facts and communicates confidence rather than presenting AI-generated analysis as certainty.

04 — Design for Action

Analysis should lead to practical next steps—investigate, validate, update, plan, or monitor.

CORE PRODUCT PHILOSOHY

AI should expand the analyst's field of view — not replace the analyst's judgment.

The product is designed as a human-in-the-loop decision-support system, where AI helps broaden analysis and surface overlooked relationships while domain experts retain responsibility for interpretation and decisions.

VALUE PROPOSITION



Business Analysts



Project Managers



Transformation Teams

Product Goals and Objectives

Primary Product Goal

Enable faster, more systematic, and more transparent change impact analysis while keeping human experts in control of final decisions.

The product aims to reduce the effort required to identify potential impacts and dependencies while improving the consistency, explainability, and actionability of early-stage change analysis.

Strategic Objectives

01 — Increase Analysis Coverage

Help users identify potential impacts across processes, people, systems, data, documentation, dependencies, and risks.

02 — Improve Analysis Consistency

Apply a structured analytical framework so that similar change requests are assessed using consistent dimensions and output formats.

03 — Reduce Investigation Effort

Use AI to accelerate the initial analysis phase and reduce repetitive manual investigation before stakeholder validation.

04 — Strengthen Decision Transparency

Provide rationale, confidence indicators, identified risks, and recommended actions so users can understand and challenge AI-generated findings.

User Outcomes

CHANGE → AI ANALYSIS → HUMAN VALIDATION → ACTION

MVP Success Criteria

Impact Coverage: Identify relevant impact areas across the CIRA dimensions

Relevance: Findings should be contextually relevant to the proposed change.

Explainability: Major findings should provide a clear rationale for why they were identified.

Actionability: Recommendations should translate identified impacts into practical next steps.

MVP success is evaluated through structured test scenarios and human review rather than AI output volume alone.

Target Users & Personas

User Persona Profiles

	Business Analyst	Project Manager	Transformation Team
Primary Goal	Build a comprehensive impact assessment	Understand change implications for delivery	Assess organization-wide change effects
Pain Point	Hidden dependencies and fragmented information	Unclear scope, risks, and affected stakeholders	Cross-functional complexity
Needs	Structured analysis and explainable findings	Risk/dependency visibility	Broader organizational perspective
Product Value	Faster and more systematic impact analysis	Better-informed planning	Structured cross-functional change intelligence

User Needs & Expectations

01 — Visibility: Understand which processes, people, systems, data, and dependencies may be affected.

02 — Structure: Assess changes through a consistent framework rather than an ad-hoc investigation.

03 — Explainability: Understand why the system identified a particular impact or risk.

04 — Confidence: Distinguish potential findings from validated information.

05 — Actionability: Translate analysis into concrete actions for investigation, validation, planning, or mitigat**ion.**

06 — Human Control: Review, challenge, and validate AI-generated findings before making decisions.

The product should give users a broader and more structured view of change—without taking decision-making authority away from them.

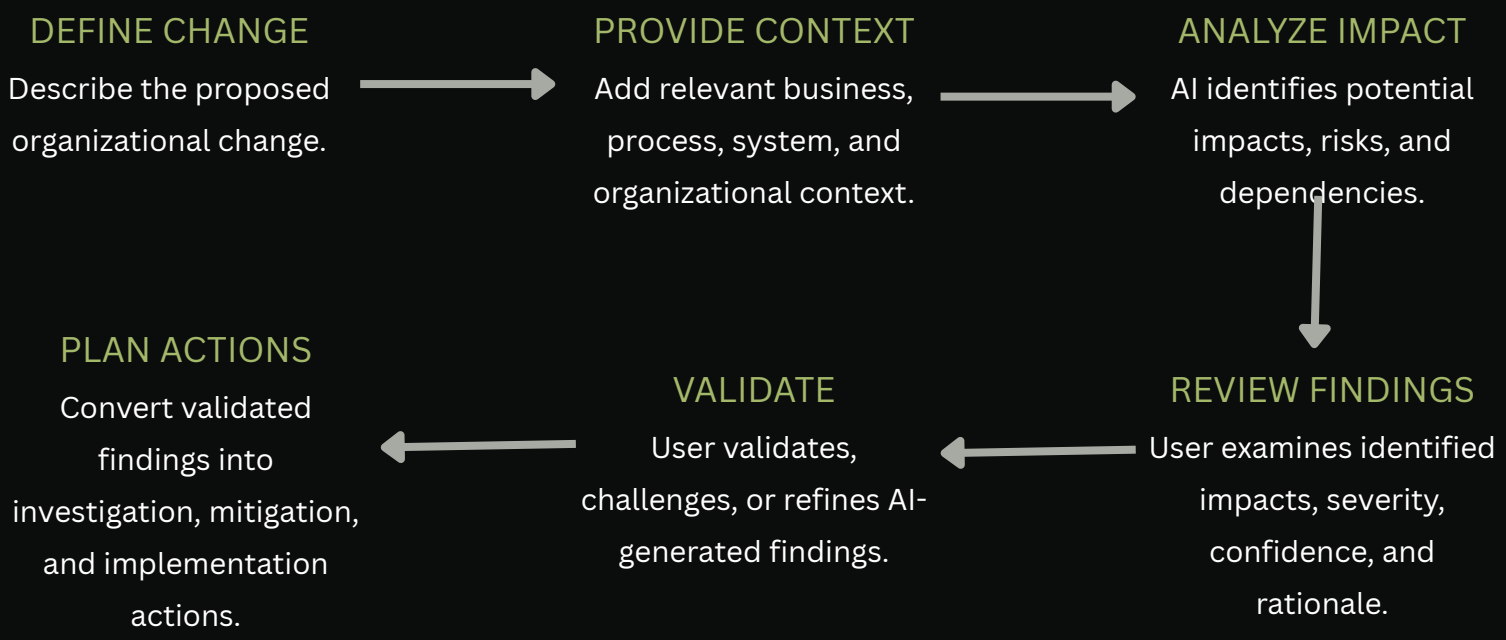
Jobs to be done & User Journey

When a change is proposed, users need to understand its potential impact across people, processes, systems, data, documentation, and dependencies, so they can identify risks, validate assumptions, and take informed action before implementation.

Change → Impact → Risk → Validation → Action

Business Analysts	Project Managers	Transformation Teams
Identify and structure potential impacts and dependencies to support comprehensive stakeholder validation.	Understand change implications, risks, and dependencies to support effective planning and execution.	Assess cross-functional effects of change to anticipate downstream consequences and prepare appropriate interventions.

Core User Journey



Human in AI Loop

AI Assists Analyze → Surface → Explain → Recommend

Human Validates Review → Validate → Challenge → Decide

AI accelerates the analysis; human experts retain responsibility for validation and decision-making.

Product Scope

The MVP focuses on analyzing proposed organizational changes and surfacing potential impacts, risks, dependencies, and recommended actions through an AI-assisted, human-in-the-loop workflow.

MVP Scope

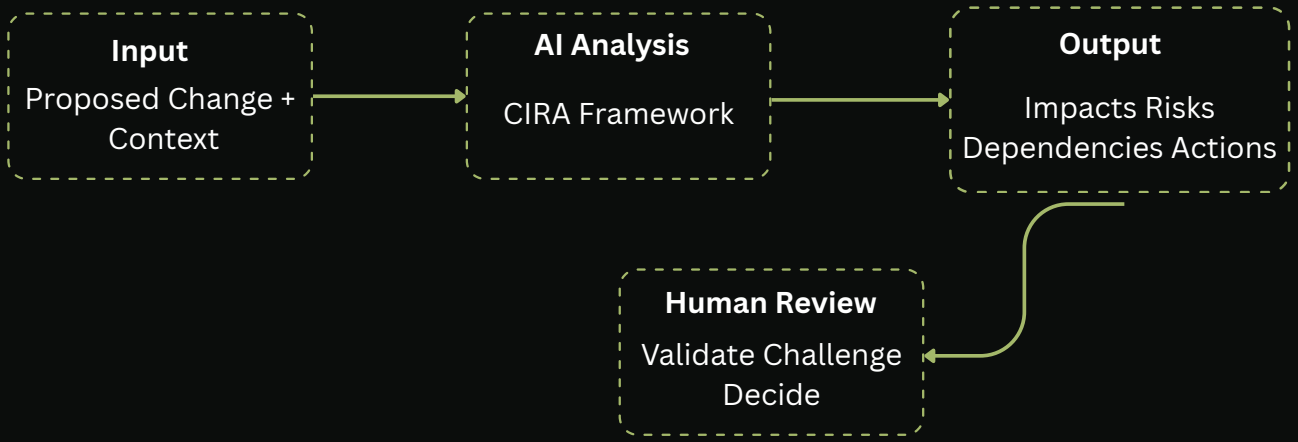
In Scope

- **Change & Context Capture**
- **AI-Assisted Impact Analysis**
- **CIRA-Based Classification**
- **Risk & Dependency Identification**
- **Severity & Confidence**
- **Explainable Findings**
- **Recommended Actions**
- **Human Review & Validation**

Out of Scope

- **Autonomous Change Approval**
- **Automatic Change Implementation**
- **Autonomous Decision-Making**
- **Guaranteed Impact Detection**
- **Enterprise-Wide Dependency Mapping**
- **Automated Stakeholder Validation**
- **Historical Change Intelligence**
- **Knowledge Graph / Advanced RAG**

MVP Boundry



The MVP ends at decision support—not decision automation.

Product Worflow

What goes into the system → what the AI does → what comes out → where the human takes over.

Step 1 — DEFINE CHANGE

User Input

- Change title
- Proposed change
- Business objective
- Affected department

Step 2 — PROVIDE CONTEXT

Contextual Information

- Current process
- Systems involved
- Teams / stakeholders
- Constraints
- Additional business context

Step 3 — AI IMPACT ANALYSIS

CIRA Analysis Engine

AI evaluates the change across:
Processes | People | Systems | Data | Documentation | Dependencies

Step 4 — RISK & DEPENDENCY ANALYSIS

The system identifies:

- Potential risks
- Upstream / downstream dependencies
- Possible consequences
- Areas requiring further investigation

Step 5 — EXPLAINABLE FINDINGS

Each significant finding includes:
Impact → Severity → Confidence → Why? → Consequences

Step 6 — RECOMMENDED ACTIONS

The system generates potential actions such as:
Investigate | Validate | Update | Mitigate | Plan | Monitor

Step 7 — HUMAN VALIDATION

The user reviews the analysis and
Confirm | Modify | Reject | Investigate

Step 8 — DECISION SUPPORT

Validated findings support:
Planning → Stakeholder Discussion → Change Decision → Implementation Preparation

Product Worflow Architecture

INPUT LAYER
Change + Context



INTELLIGENCE LAYER
Context Processing → CIRA Analysis → Risk & Dependency Analysis → Explainability



DECISION-SUPPORT LAYER
Findings → Recommendations → Human Validation → Action

Human Control Point

Human validation is a deliberate control point in the workflow—not an exception or fallback.

AI Analysis → Human Review → Validated Assessment

User Experince & Screens

User Needs & Expectations

The user experience shall enable users to move from a proposed organizational change to a structured, explainable, and reviewable impact assessment with minimal friction.
The interface shall present the overall assessment first, followed by detailed findings, risks, dependencies, and recommended actions. Users shall be able to investigate individual findings without losing sight of the overall impact.
The experience shall prioritize clarity, explainability, confidence, and human control. AI-generated findings must be presented as potential impacts until reviewed and validated by an authorized user.

UX Design Principles

01 — Structured Analysis: The interface shall organize findings into consistent impact categories, allowing users to scan, compare, and investigate results systematically.

02 — Progressive Disclosure: The interface shall present high-level findings first and allow users to access detailed rationale, consequences, dependencies, and recommendations when needed.

03 — Explain ability: Significant findings shall include a clear rationale explaining why the impact was identified. Confidence indicators shall help users understand uncertainty without implying guaranteed accuracy.

04 — Human Validation: Users shall be able to review, validate, modify where supported, or reject individual findings before relying on them for change planning.

05 — Action-Oriented Experience: Findings and risks shall be connected to practical next steps so users can move from analysis to investigation, validation, mitigation, and planning.

UX Design Principles

Screen 01 — Dashboard

Objective: Provide a central entry point for initiating and accessing change impact assessments.

Functional Requirements

- The system shall allow users to initiate a new change impact analysis.
- The dashboard shall display previously created assessments when analysis history is supported.
- Each saved assessment shall display its title, creation or update date, and current status.
- Users shall be able to reopen an existing assessment when persistence is implemented.
- The interface shall provide a clear entry point to begin a new assessment.

Primary Action: Analyze New Change.

Acceptance Criteria: A user can initiate a new assessment without navigating through unnecessary steps.

Screen 02 — Change Input

Objective: Capture the proposed change and sufficient business context to support meaningful analysis.

Required Input

- Change title
- Proposed change description
- Business objective

Optional Context

- Affected department or function
- Current-state process
- Systems involved

- Teams or stakeholders
- Constraints and assumptions
- Additional business context

Functional Requirements

- The system shall validate required fields before submitting an analysis.
- Users shall be able to provide optional context to improve the relevance of the assessment.
- The system shall communicate missing required information clearly.
- The system shall submit the change description and supplied context to the analysis service.
- The interface shall indicate when analysis is in progress and prevent accidental duplicate submissions where appropriate.

Acceptance Criteria: Users can submit a valid change request, receive clear validation messages for incomplete input, and understand when processing is underway.

Screen 03 — Impact Analysis Overview

Objective: Provide a high-level view of the potential organizational impact before users investigate individual findings.

Information Displayed

- Overall impact assessment
- Impact categories
- Number of identified findings
- High-severity findings
- Key risks and dependencies
- Overall assessment confidence, where supported

Impact Categories

- Processes
- People
- Systems
- Data
- Documentation
- Dependencies

Functional Requirements

- The system shall organize findings by impact category.
- Each finding shall display its severity and confidence indicators when those assessments are available.
- Users shall be able to navigate from the overview to the relevant detailed findings.
- The interface shall distinguish an overall assessment from the severity of individual findings.
- The system shall display an appropriate empty or incomplete state if no findings are returned or the analysis cannot be completed.

Acceptance Criteria:
Users can identify the principal areas of potential impact and navigate to supporting findings without interpreting an unstructured AI response.

Screen 04 — Impact Details & Finding Review

Objective: Enable users to understand, assess, and validate individual AI-generated findings

Each Finding Shall Include

- Impacted area
- Impact description
- Severity
- Confidence indicator
- Rationale explaining why the impact was identified
- Potential consequences
- Related risks or dependencies, where available
- Recommended action

Functional Requirements

- The system shall display findings in a consistent, structured format.
- Users shall be able to mark each finding as Validated, Needs Review, or Not Relevant.
- The system shall distinguish AI-generated content from user-validated information.
- Users shall be able to inspect the rationale and supporting context associated with a finding.
- The system shall retain the user's validation status when the assessment is saved.
- The system shall not automatically treat an AI-generated finding as validated.

Acceptance Criteria: Users can review each finding, understand the reason behind it, and record its validation status without ambiguity.

Screen 05 — Risks & Dependencies

Objective: Help users understand potential failure points and relationships that could influence the proposed change.

Information Displayed

- Potential operational risks
- Process dependencies
- System and integration dependencies
- Data dependencies
- Cross-functional dependencies
- Potential downstream consequences
- Areas requiring further investigation

Functional Requirements

- The system shall present identified risks and dependencies separately from general impact findings.
- Where possible, each risk or dependency shall reference its associated finding.
- The interface shall distinguish identified relationships from assumptions requiring validation.
- The system shall identify areas where insufficient context prevents a reliable assessment.
- The interface shall not imply that all organizational dependencies have been identified.

Acceptance Criteria:
Users can understand the principal risks and dependencies, trace them to related findings where possible, and identify items requiring further investigation.

Screen 06 — Recommendations & Action Planning

Objective: Translate potential impacts and risks into practical next steps.

Supported Action Types
Investigate | Validate | Update | Mitigate | Plan | Monitor

Functional Requirements

- The system shall generate recommendations relevant to identified impacts or risks.
- Where possible, each recommendation shall reference the finding or risk that prompted it.

- Recommendations shall describe an actionable next step rather than provide only generic advice.
- Users shall be able to review recommendations before incorporating them into their change plan.
- Where action tracking is implemented, users shall be able to record the action status.
- Acceptance Criteria: Users can identify what needs to be done next and understand how the recommended action relates to the assessment.

Screen 07 — Decision-Support Report

Objective: Consolidate the assessment into a structured report for stakeholder discussions and change planning.

Report Structure

- Change Overview
- Overall Impact Assessment
- Identified Impacts
- Risks
- Dependencies
- Recommended Actions
- Confidence and Uncertainty
- Human Validation Status

Functional Requirements

- The system shall present the assessment in a consistent report structure.
- The report shall distinguish AI-generated findings from human-validated findings.
- The report shall include relevant rationale, risks, dependencies, and recommendations.
- The report shall communicate material gaps or uncertainties in the available context.
- The report shall support a practical sharing or export mechanism when that capability is implemented.

Acceptance Criteria: A user can review the assessment as a coherent decision-support document and identify which findings have been validated and which still require investigation.

Information Hierarchy

The experience shall follow this analytical sequence:
CHANGE → IMPACT → WHY → RISK → ACTION → VALIDATION

Each stage shall answer a specific user question:
Change: What is being proposed?
Impact: What could be affected?
Why: Why was the potential impact identified?
Risk: What could go wrong?
Action: What should be investigated or done next?
Validation: What has been reviewed or confirmed by a human?

This hierarchy shall remain consistent across the impact overview, detailed findings, and decision-support report.

System States & Error Handling

The experience shall be evaluated against the following criteria:

- **Ease of Use:** Users can submit a valid change request without unnecessary configuration.
- **Clarity: Users** can understand the overall assessment and locate significant findings.
- **Explainability:** Users can understand why important impacts were identified.
- **Transparency:** Users can distinguish potential findings from human-validated findings.
- **Actionability:** Users can connect identified impacts and risks to practical next steps.
- **Error Recovery:** Users receive understandable guidance when input is invalid or analysis fails.
- **Human Control:** Users can review and challenge AI-generated findings before relying on them for decisions.

These criteria shall be assessed through representative user scenarios and structured evaluation. Numerical performance targets shall be established in the Evaluation Framework after the initial baseline is measured.

Core UX Principle

The interface should make complexity easier to navigate—not hide complexity from the user.

CIRA Framework

Framework Overview

The Change Impact & Risk Analysis (CIRA) Framework is the core analytical framework of the AI Change Impact Analyzer. It provides a structured approach for examining proposed organizational changes, identifying potential impacts, assessing associated risks, and recommending appropriate next steps.

The framework ensures that the AI evaluates changes across consistent impact dimensions rather than producing an unstructured response.

CIRA consists of four stages: **Change, Impact, Risk, and Action.**

CIRA Stages

C — Change

Objective: Establish what is changing and why.

The system shall interpret the proposed change using the information supplied by the user, including the change description, business objective, current-state process, affected department, systems, stakeholders, and relevant constraints.

The system shall identify the primary change area and use the available context to guide subsequent analysis.

Expected Output:

- Change summary
- Business objective
- Primary change area
- Relevant context and assumptions
- Identified information gaps

I — Impact

Objective: Identify areas that could be affected directly or indirectly by the proposed change.

The system shall assess potential impacts across six predefined dimensions:

1. Processes: Business workflows, procedures, approvals, handoffs, and operational activities.

2. People: Roles, responsibilities, teams, stakeholders, training needs, and adoption considerations.

3. Systems: Applications, integrations, technical components, and system interactions.

4. Data: Data inputs, outputs, ownership, quality, access, and information flows.

5. Documentation: Standard operating procedures, policies, business requirements, training materials, and process documentation.

6. Dependencies: Relationships between processes, teams, systems, data, and other activities that may be affected by the change.

The system shall identify potential impacts and explain the relationship between the proposed change and each finding. Where evidence or context is insufficient, the finding shall be presented as a hypothesis requiring validation.

Expected Output:

- Impacted area
- Impact description
- Impact category
- Potential consequences
- Supporting rationale
- Related dependencies

R — Risk

Objective: Identify potential adverse outcomes and areas requiring further investigation.

The system shall evaluate the potential risks associated with identified impacts and dependencies. Risk analysis shall consider the nature of the potential consequence, the context provided, and the uncertainty associated with the finding.

Potential risk categories include:

- Operational disruption
- Process failure
- Data quality or integrity issues
- System integration failures
- Compliance or policy gaps
- Stakeholder resistance or adoption challenges
- Incomplete or unvalidated dependencies

Each significant risk shall include a description, its potential consequence, an explanation of why it matters, and a recommended mitigation or investigation action where appropriate.

Severity Assessment

- **Critical:** Potential for severe business disruption or significant adverse consequences requiring urgent expert assessment.
- **High:** Potential for substantial operational, technical, or business impact.
- **Medium:** Potential for a material but more contained impact.
- **Low:** Potential for limited impact that may require routine follow-up.

Severity shall reflect the potential consequence, not the AI's confidence that the risk will occur. Where sufficient context is unavailable, the system shall communicate the uncertainty rather than imply a definitive assessment.

Confidence Assessment

- **High:** The finding is strongly supported by the supplied context and a clear rationale.
- **Medium:** The finding is plausible, but some assumptions or information gaps remain.
- **Low:** The finding is tentative and requires additional context or expert validation.

Confidence represents the system's assessment of support for a finding; it shall not be presented as a statistically calibrated probability unless such calibration has been implemented and evaluated.

Expected Output:

- Risk description
- Severity
- Confidence
- Potential consequence
- Rationale and assumptions
- Recommended mitigation or validation action

A — Action

Objective: Translate potential impacts and risks into practical next steps.

The system shall generate recommendations that help users investigate, validate, mitigate, or prepare for the proposed change.

Supported action types include:

- **Investigate:** Gather missing information or examine an uncertain impact.
- **Validate:** Confirm assumptions with relevant stakeholders or subject-matter experts.

- **Update:** Revise affected processes, policies, data definitions, or documentation.
- **Mitigate:** Reduce the likelihood or consequence of an identified risk.
- **Plan:** Prepare implementation, communication, training, or testing activities.
- **Monitor:** Identify areas requiring observation after implementation.

Recommendations shall be relevant to the associated finding or risk wherever possible. The system shall provide decision support only; users remain responsible for determining which actions are appropriate and approving implementation decisions.

Expected Output:
- Recommended action
- Reason for the action
- Related impact or risk
- Suggested validation area or stakeholder, where identifiable
- Priority for follow-up, where supported

Impact Finding Structure

Each identified impact shall follow a consistent structure so that findings can be reviewed, compared, validated, and used by downstream application components.
Each finding shall contain:

1. **Finding ID**: A unique identifier within the assessment.
2. **Category:** One of the six defined impact dimensions.
3. **Affected Area**: The process, role, system, data, document, or dependency potentially affected.
4. **Impact Description**: What could change or be affected.
5. **Severity:** The assessed level of potential consequence.
6. **Confidence**: The assessed strength of support for the finding.
7. **Rationale**: Why the system identified the potential impact.
8. **Potential Consequences**: What may happen if the impact is not addressed.
9. **Related Risks and Dependencies**: Associated risks or relationships, where identified.
10. **Recommended Action**: A practical next step.
11. **Validation Status**: The current review status of the finding.

The initial validation statuses shall be Needs Review, Validated, and Not Relevant. A finding shall default to Needs Review until a user changes its status.

Framework Application Rules

The system shall apply the CIRA Framework consistently to each valid change request.

- It shall examine all six impact dimensions, even when no potential impact is identified in a particular dimension.
- It shall distinguish identified impacts from assumptions and information gaps.
- It shall provide a rationale for significant findings.
- It shall assess severity separately from confidence.
- It shall connect risks and recommendations to relevant findings wherever possible.
- It shall avoid inventing organizational facts, system relationships, or stakeholder responsibilities not supported by the supplied context.
- It shall allow users to review and challenge findings before relying on them.
- It shall communicate that the framework may not identify every possible impact or dependency.

Example Application

Proposed Change: Automate the vendor onboarding process.

Change: Replace selected manual vendor onboarding activities with an automated workflow.

Potential Impacts:

- **Process:** Vendor registration and approval procedures may need to change.
- **People:** Procurement and Finance teams may need updated responsibilities or training.
- **Systems:** The workflow may require integration with an enterprise resource planning system.
- **Data:** Vendor master data may require validation and consistent field definitions.
- **Documentation:** Onboarding procedures and approval policies may need updating.
- **Dependencies:** Existing approval hierarchies and compliance checks may affect the new workflow.

Potential Risk: Incorrect integration or incomplete validation could result in inaccurate vendor records or interrupted onboarding.

Recommended Action: Validate the approval hierarchy, vendor data requirements, and system integration dependencies with the relevant business and technical stakeholders before implementation

Human Validation: The analyst reviews the findings, confirms which relationships apply to the organization, and identifies any additional impacts.

This example illustrates possible findings rather than confirmed facts about a specific organization.

Framework Success Criteria

The CIRA Framework shall be evaluated on its ability to:

- Cover all six impact dimensions consistently.
- Generate contextually relevant potential impacts.
- Provide understandable rationales.
- Distinguish severity from confidence.
- Identify meaningful risks and dependencies.
- Produce actionable recommendations linked to findings.
- Communicate uncertainty and avoid unsupported organizational claims.
- Support human review and validation.

The framework's performance shall be measured through structured test scenarios and human evaluation, as defined in the Evaluation Framework.

Core Principle: CIRA structures the analysis; AI assists with identifying potential implications; human experts validate findings and retain decision-making authority

AI Requirement & Architecture

Objective

The objective of the AI architecture is to transform a proposed organizational change and its supporting context into a structured, explainable, and reviewable impact assessment.

The system shall use a Large Language Model (LLM) to interpret the change request, apply the CIRA Framework, identify potential impacts and risks, and generate recommended actions.

The AI component shall operate within a controlled application architecture that validates model outputs, handles failures, and preserves human oversight.

The architecture shall separate the user interface, application logic, AI integration, and data persistence so that each component can be developed, tested, and maintained independently.

Proposed Technology Stack

The following technologies are proposed for the initial implementation. Final selections shall be confirmed during technical design.

Frontend — Next.js, React and TypeScript

The frontend provides the user-facing experience, including change input forms, impact summaries, detailed findings, validation controls, and decision-support reports.
React supports reusable interface components, while TypeScript helps identify data and type errors during development.

Backend — Next.js Server-Side API

The backend receives requests from the frontend, validates input, calls the AI service, processes responses, and returns structured results.
Keeping this logic on the server prevents AI API credentials from being exposed in browser-side code.

For the MVP, a separate backend framework is not required unless implementation complexity justifies it.

AI Model — Large Language Model API
An LLM interprets the proposed change and generates potential impacts, risks, dependencies, explanations, and recommended actions.

The model shall be accessed through a server-side integration. The application shall not depend on a particular development interface to execute its analysis.

Output Validation — Zod or JSON Schema
AI responses may be incomplete or contain unexpected fields. A validation layer checks that the returned data follows the application's expected structure before it is displayed or saved.

Invalid responses shall trigger controlled error handling or a bounded retry where appropriate.

Database — PostgreSQL through Supabase (if persistence is included in the MVP)
A database can store change requests, generated findings, user validation statuses, and assessment history.
Supabase provides a managed PostgreSQL option and can support authentication and database access controls if those capabilities are required.

If the initial MVP is a single-user demonstration, persistence and authentication may be introduced incrementally rather than delaying the core analysis workflow.

Version Control — Git and GitHub
GitHub shall host the application source code, technical documentation, test cases, and project history. This supports collaboration, change tracking, and reproducible development.

High-Level System Architecture

The product shall follow a modular architecture consisting of five logical layers.

Layer 1 — User Interface

Captures the proposed change and context, presents the assessment, and enables users to review findings and record validation decisions.

Layer 2 — Application Backend

Validates user input, manages analysis requests, coordinates the AI service, handles errors, and prepares the response for the frontend.

Layer 3 — AI Analysis Engine

Uses the configured LLM and structured instructions to analyze the change across the six CIRA impact dimensions and produce potential impacts, risks, dependencies, and recommendations.

Layer 4 — Output Validation & Normalization

Checks the AI response against the expected data structure, verifies required fields and allowed values, and normalizes the result before presentation or persistence.

Structural validation does not prove that an AI-generated finding is factually correct. Human review and evaluation remain necessary.

Layer 5 — Data Persistence

Where enabled, stores the assessment, findings, and human validation status so users can retrieve previous analyses and preserve review decisions.

Architecture Flow

User Input → Frontend → Backend Validation → AI Model API → Output Validation → Assessment Results → Human Review → Optional Database Persistence

AI Analysis Requirements

The AI component shall satisfy the following requirements.

AI-REQ-01 — Contextual Interpretation

The system shall interpret the proposed change using the description and contextual information supplied by the user.

AI-REQ-02 — CIRA-Based Analysis

The system shall generate potential impacts across Processes, People, Systems, Data, Documentation, and Dependencies.

AI-REQ-03 — Risk Identification

The system shall identify plausible risks associated with the proposed change and explain their potential consequences.

AI-REQ-04 — Explainable Findings

Each significant finding shall include a rationale explaining the relationship between the proposed change and the potential impact.

AI-REQ-05 — Structured Output
The AI service shall return data in a predefined machine-readable format suitable for validation and rendering by the application.

AI-REQ-06 — Severity and Confidence
The system shall assess potential severity separately from confidence and shall not represent confidence labels as calibrated probabilities unless calibration has been implemented and evaluated.

AI-REQ-07 — Action Recommendations
The system shall generate relevant next steps linked to identified impacts or risks wherever possible.

AI-REQ-08 — Uncertainty Handling
The system shall communicate material information gaps and distinguish assumptions or potential impacts from human-validated findings.

AI-REQ-09 — Unsupported Claims
The system shall avoid presenting unknown organizational facts, undocumented integrations, or unverified dependencies as established facts.

AI-REQ-10 — Human Oversight
The system shall not autonomously approve, reject, or implement organizational changes.

Structured AI Response

The AI service shall return an assessment object containing the overall assessment, potential findings, risks, dependencies, and recommended actions.

The following illustrates the proposed logical response structure:

```json
{
  "assessment": {
    "changeTitle": "Automate vendor onboarding",
    "overallImpact": "HIGH",
    "summary": "Potential cross-functional workflow changes require review.",
    "limitations": [
      "System integration details have not been confirmed."
    ]
  },
  "impacts": [
    {
      "id": "IMP-001",
      "category": "PROCESS",
      "affectedArea": "Vendor onboarding",
      "description": "Approval steps may need to change.",
      "severity": "HIGH",
      "confidence": "MEDIUM",
      "rationale": "Automation may alter existing approval handoffs.",
      "validationStatus": "NEEDS_REVIEW",
      "actions": [
        "Validate the approval workflow with Procurement."
      ]
    }
  ],
  "risks": [],
  "dependencies": [],
  "recommendations": []
}
```


This is an illustrative contract, not a claim that the implementation already returns this response. The final schema shall be defined in code and version-controlled alongside the application.

The application shall reject or safely handle responses that fail schema validation rather than rendering malformed data as a completed assessment.

AI Processing Workflow

The AI analysis process shall follow these stages:

1. **Receive Input:** Accept the proposed change and contextual information.
2. **Validate Request:** Check required fields, input types, and request limits.
3. **Prepare Context:** Organize the information into a consistent representation for analysis.
4. **Apply CIRA Instructions:** Instruct the model to examine all six impact dimensions and identify potential risks and actions.
5. **Generate Structured Findings:** Request an assessment using the predefined output schema.
6. **Validate Response:** Check schema compliance and required values.
7. **Handle Incomplete Results:** Retry within defined limits or return a clear error when the result cannot be used.
8. **Present Assessment:** Display the validated response for human review.
9. **Record Review Decisions:** Persist validation status when storage is enabled.

Prompt Design & Control

The application shall use version-controlled prompts to guide the AI model's analysis.
Prompt instructions shall define the model's role, the CIRA Framework, required output fields, severity and confidence definitions, uncertainty handling, and prohibited behavior.

The system shall instruct the model to:

- Evaluate all six impact dimensions.
- Explain the rationale for significant findings.
- Separate potential impacts from confirmed information.
- Identify assumptions and missing context.
- Avoid inventing organization-specific facts.
- Link recommendations to relevant findings where possible.
- Return the expected structured response.
- Preserve human responsibility for decisions.

Prompt changes shall be evaluated against a consistent test dataset before being accepted into a release.
Prompts improve consistency but do not guarantee factual accuracy or eliminate hallucinations.

Security, Privacy & Data Handling

The system shall follow these baseline requirements:

- AI API keys and other secrets shall be stored in server-side environment variables or an appropriate secret-management service.
- Secrets shall not be committed to GitHub or exposed in frontend code.
- User inputs shall be validated before processing.
- Errors and application logs shall avoid unnecessarily recording sensitive business information.
- Data persistence, if enabled, shall use appropriate access controls.
- The system shall communicate what information is sent to the configured AI provider.
- Sensitive organizational information shall not be sent to external AI services without an appropriate data-handling decision.
- The application shall use secure transport and follow the selected hosting and database providers' security recommendations.

Authentication, authorization, retention policies, and deletion controls shall be implemented according to the agreed deployment context and MVP scope.

Architecture Evolution

The architecture shall support future improvements without requiring a complete redesign.

Version 1 — Structured LLM Analysis

Analyze the user-provided change and context using a predefined framework, controlled prompts, and structured output validation.

Version 2 — Context-Aware Analysis

Introduce retrieval-augmented generation (RAG) if access to relevant organizational documents is required. Retrieval can supply approved context to the model, but retrieved information must be relevant, permission-controlled, and traceable to its source.

Version 3 — Advanced Dependency Intelligence

Explore knowledge graphs, explicit dependency relationships, and historical change data to improve cross-functional impact analysis.

These are roadmap directions, not existing capabilities. The initial MVP shall not claim document retrieval, a knowledge graph, or automated enterprise dependency discovery unless those features have actually been implemented and evaluated.

Technical Acceptance Criteria

The AI architecture shall be considered ready for MVP evaluation when:

- A valid change request can be submitted through the application.
- The backend can invoke the configured AI model without exposing credentials to the browser.
- The response follows the defined output schema.
- Findings cover all six CIRA dimensions in a consistent structure, including when a dimension has no identified impact.
- Significant findings include rationale, severity, and confidence where applicable.
- Invalid responses and AI service failures are handled safely.
- The user can review and validate findings without the system treating them as confirmed automatically.
- Representative test cases can be executed and their results recorded.
- The codebase includes setup instructions and documented configuration requirements.

Architecture Principle: The LLM generates potential analytical findings; the application validates the response structure; the human user remains responsible for validating findings and making organizational decisions.

Functional Requirements

Purpose

This section defines the functional requirements of the AI Change Impact Analyzer. It describes what the system must do, how users interact with it, and how the system should respond to different inputs and situations.

Each requirement has a unique identifier, priority, and acceptance criteria. These requirements will guide development and help determine whether the product behaves as expected.

Requirement Priorities

Requirements are classified into three priority levels:

- **P0 —** Must Have: Essential for the core product to function.
- **P1 —** Should Have: Important improvements that strengthen usability and analysis quality.
- **P2 —** Could Have: Optional enhancements that can be considered after the initial version.

The initial release should prioritize a working change-analysis workflow over additional dashboards, integrations, and reporting features.

Change Input and Context Management

FR-001: Enter a Proposed Change

Priority: P0

The system shall allow users to describe a proposed organizational, operational, process, or technology change.

Acceptance criteria:

- Users can enter a change title and description.
- The system accepts readable text describing the proposed change.
- The system prevents analysis when the required change description is missing.
- The system displays a clear validation message when the input is incomplete.

FR-002: Provide Organizational Context

Priority: P0

The system shall allow users to provide relevant context that may influence the impact analysis.

Context may include affected departments, existing workflows, systems, stakeholders, objectives, and known constraints.

Acceptance criteria:

- Users can provide additional context alongside the change description.
- The system uses the supplied context when generating findings.
- Missing context is identified where it could affect the reliability of the analysis.
- The system does not treat unspecified organizational details as confirmed facts.

FR-003: Validate Analysis Inputs

Priority: P0
The system shall check the completeness and validity of the submitted information before requesting AI analysis.

Acceptance criteria:

- Required fields are validated before submission.
- Empty or invalid submissions are rejected with understandable messages.
- Input length and supported formats are controlled by application rules.
- Validation errors do not trigger unnecessary AI requests.

AI-Powered Change Impact Analysis

FR-004: Generate a Structured Impact Assessment

Priority: P0
The system shall use an AI model to generate a structured assessment based on the proposed change and the context supplied by the user.

Acceptance criteria:

- The assessment includes a change summary and potential overall impact level.
- Findings are organized into the CIRA framework.
- The response follows the application's defined output schema.
- The system handles unsuccessful or invalid AI responses without presenting them as successful assessments.

FR-005: Analyze All Six Impact Dimensions

Priority: P0

The system shall examine the proposed change across the six CIRA impact dimensions:

1. Processes
2. People
3. Systems
4. Data
5. Documentation
6. Dependencies

Acceptance criteria:

- All six dimensions are considered during analysis.
- Relevant potential impacts are presented under the appropriate dimension.
- Dimensions with no identified impact are distinguished from dimensions that could not be assessed due to insufficient information.
- The system does not invent findings simply to populate every dimension.

FR-006: Explain Potential Impacts

Priority: P0

The system shall provide a clear explanation for each identified potential impact.

Acceptance criteria:

- Each finding includes a description of the affected area.
- Each finding includes a rationale explaining why the change may affect that area.
- Potential consequences are included when they can reasonably be inferred from the available context.
- Unsupported assumptions are identified as uncertain rather than presented as established facts.

FR-007: Classify Impact Severity

Priority: P0

The system shall assign a proposed severity level to each impact finding using the defined CIRA severity scale.

The severity levels are Critical, High, Medium, and Low.

Acceptance criteria:

- Each impact finding receives a severity classification when sufficient information is available.
- The meaning of each severity level is defined consistently.
- The system provides a rationale for the classification.
- Users are informed when severity cannot be assessed reliably from the available information.

FR-008: Indicate Confidence and Uncertainty

Priority: P0

The system shall indicate how strongly each finding is supported by the supplied context.

Acceptance criteria:

- Findings include High, Medium, or Low confidence where applicable.
- Confidence is presented separately from severity.
- The system explains important information gaps or assumptions.
- Confidence labels are not presented as statistically calibrated probabilities unless a validated calibration method has been implemented.

Risk and Dependency Identification

FR-009: Identify Potential Risks

Priority: P0

The system shall identify potential risks arising from the proposed change and its associated impacts.

Potential risk areas include operational disruption, process failures, data issues, integration failures, compliance gaps, and adoption challenges.

Acceptance criteria:

- Identified risks include a description and relevant context.
- Risks are linked to related impacts where a meaningful relationship can be established.
- Risk severity is assigned when sufficient information is available.
- The system distinguishes potential risks from confirmed issues.

FR-010: Identify Dependencies

Priority: P0
The system shall identify known or potential dependencies that may affect implementation or successful adoption of the change.

Acceptance criteria:

- Dependencies are described in understandable language.
- Relevant relationships with impacts or risks are displayed where available.
- Unconfirmed dependencies are clearly marked for validation.
- The system does not claim to have discovered every organizational dependency.

Recommendations and Human Review

FR-011: Generate Actionable Recommendations

Priority: P0
The system shall suggest practical next steps to investigate, validate, mitigate, or manage identified impacts and risks.

Acceptance criteria:

- Recommendations relate to identified findings or information gaps.
- Each recommendation explains the action to consider.
- Recommendations identify a suggested responsible function or stakeholder when the available context supports it.
- Recommendations are presented as suggestions, not automatically approved decisions.

FR-012: Support Finding Validation

The system shall allow users to review and classify individual findings.
The available review statuses shall include:

- Needs Review
- Validated
- Not Relevant

Acceptance criteria:

- Each finding displays its current review status.
- Findings default to Needs Review.
- Users can change the status of a finding.
- The interface clearly distinguishes AI-generated findings from findings reviewed by a human.

For the initial release, review status may be maintained within the active assessment session. Persistent storage across sessions is a separate requirement unless included in the agreed MVP scope.

FR-013: Preserve Human Decision Authority

Priority: P0
The system shall support human judgment throughout the impact analysis process.

Acceptance criteria:

- AI-generated findings are presented for review rather than automatic approval.
- Users can identify findings that require further investigation.
- The system does not independently approve or implement organizational changes.
- The interface communicates that the assessment supports, rather than replaces, professional judgment.

Results Presentation and System Feedback

FR-014: Display an Organized Assessment

Priority: P0
The system shall present analysis results in a structured and readable format.

Acceptance criteria:

- The change summary, impacts, risks, dependencies, and recommendations are clearly separated.
- Findings can be understood without requiring users to interpret raw JSON.
- Severity and confidence are visually distinguishable.
- Relevant rationale and uncertainty information are accessible to the user.

FR-015: Display Analysis Progress

Priority: P1
The system should inform users when an analysis request is being processed.

Acceptance criteria:

- A loading or progress state appears after submission.
- Users receive a clear indication when the assessment is ready.
- Users are informed if the request fails or takes longer than the defined response threshold.
- Duplicate submissions are controlled where appropriate.

FR-016: Handle Incomplete Results and Errors

Priority: P0

The system shall handle insufficient context, invalid AI output, service failures, and other unsuccessful analysis conditions.

Acceptance criteria:

- Users receive a clear explanation when an assessment cannot be completed.
- Invalid or incomplete output is not silently treated as a complete assessment.
- The system identifies important limitations in the results.
- Users are guided to provide additional context or retry when appropriate.
- Technical errors do not expose API keys, confidential system details, or unnecessary diagnostic information.

Optional and Future Functional Requirements

The following features may improve the product but should not be treated as committed MVP capabilities unless explicitly included in the approved scope.

FR-017: Save and Reopen Assessments

Priority: P2

The system could allow users to save completed assessments and return to them later.

Acceptance criteria, if implemented:

- Saved assessments can be retrieved by authorized users.
- Assessment details and review statuses are preserved.
- Access controls prevent unauthorized viewing.

FR-018: Export Assessment Reports

Priority: P2

The system could allow users to export assessments into a shareable report format.

Acceptance criteria, if implemented:

- The report includes the change summary, impacts, risks, dependencies, recommendations, and limitations.
- Exported content reflects the latest available assessment and review statuses.
- The report clearly distinguishes AI-generated findings from human-validated findings.

FR-019: Track Actions and Ownership

Priority: P2
The system could support action owners, target dates, and action statuses.

Acceptance criteria, if implemented:

- Actions can be assigned to responsible stakeholders.
- Target dates and progress statuses can be recorded.
- Action tracking is clearly distinguished from recommendations that have not yet been approved.

Functional Acceptance Principle

A functional requirement is considered satisfied only when its stated behavior can be demonstrated and its acceptance criteria can be tested.

Requirements that depend on external services, database persistence, authentication, or export functionality must be validated against the actual implemented system.

The product shall not claim that a feature is available merely because it appears in the PRD or in an AI-generated response.

Core principle: Every requirement must describe observable product behavior—not just an intended capability.

FR-017: Save and Reopen Assessments

Priority: P2
The system could allow users to save completed assessments and return to them later.

Acceptance criteria, if implemented:

- Saved assessments can be retrieved by authorized users.
- Assessment details and review statuses are preserved.
- Access controls prevent unauthorized viewing.

Non-Functional Requirement

Purpose

Non-functional requirements define the quality standards and operational expectations of the AI Change Impact Analyzer. They ensure that the application is usable, reliable, secure, maintainable, and suitable for its intended purpose.

Since the product uses AI to support organizational decision-making, particular attention must be given to output quality, transparency, data protection, and the limitations of AI-generated findings.

These requirements will guide technical implementation, testing, and product evaluation.

Performance

NFR-001: Analysis Response Time

Priority: P0

The system should return an impact assessment within a reasonable period under normal operating conditions.

Target:

- Target a response time of 30 seconds or less for at least 90% of standard analysis requests under the defined test conditions.
- Display a loading state while an assessment is being generated.
- Handle requests that exceed the target through appropriate timeout and error handling.

Measurement: Measure end-to-end response time from submission to display of the result across a representative test set.

The target is a proposed benchmark and must be verified against the selected AI provider, model, input size, and deployment environment.

NFR-002: Interface Responsiveness

Priority: P0

The interface shall remain responsive and readable during normal user interactions.

Acceptance criteria:

- Input fields and navigation controls respond without unnecessary delay.

- Long assessments remain readable and navigable.
- The interface provides feedback during time-consuming operations.
- AI processing does not unnecessarily block unrelated interface interactions.

Reliability and Availability

NFR-003: Graceful Failure Handling

Priority: P0

The system shall handle AI provider failures, network interruptions, timeouts, and unexpected responses without crashing the application.

Acceptance criteria:

- Errors are communicated in clear, user-friendly language.
- Failed requests are not presented as successful assessments.
- Users can retry an unsuccessful request when appropriate.
- Sensitive technical details are excluded from user-facing error messages.

NFR-004: Output Integrity

Priority: P0

The system shall preserve the structural integrity of assessment results.

Acceptance criteria:

- AI responses are checked against the expected output schema.
- Invalid fields, missing required values, and unsupported classifications are handled according to defined validation rules.
- Invalid results are not silently accepted as complete.
- Any recovery or correction attempt is subject to validation before the results are displayed

NFR-005: Service Availability

Priority: P1

The application should remain accessible during its intended operating periods, subject to hosting and third-party service availability.

Acceptance criteria:

- Hosting and AI-provider failures are handled gracefully.
- External service dependencies are documented.

- Availability can be monitored through deployment or service-level monitoring where configured.
- An uptime percentage should be established after the hosting environment and expected usage are defined.

Security and Data Protection

NFR-006: API Key Protection

Priority: P0

All private API keys and secrets shall be stored securely on the server side.

Acceptance criteria:

- AI provider credentials are never embedded in client-side code.
- Secrets are not committed to public source-control repositories.
- Environment variables or an appropriate secrets-management mechanism are used.
- Credentials can be rotated without redesigning the application.

NFR-007: Input and API Security

Priority: P0

The system shall validate incoming requests and protect its application endpoints against common misuse.

Acceptance criteria:

- Input types, lengths, and required fields are validated on the server.
- The backend enforces permitted request formats.
- Appropriate rate limits or abuse controls are applied to exposed endpoints.
- Errors do not reveal secrets or unnecessary internal implementation details.

NFR-008: Data Privacy

Priority: P0

The system shall handle organizational information according to the application's disclosed data practices.

Acceptance criteria:

- Users are informed when submitted information is processed by an external AI provider.
- Only information necessary for the analysis is sent to the model.

- Sensitive organizational information is not submitted to an external provider without an appropriate data-handling decision.
- Logs avoid unnecessary storage of confidential input and full AI responses.
- Data retention and deletion behavior is documented for the deployed version.

NFR-009: Access Control

Priority: P1 — Conditional on persistent user data

If user accounts or saved assessments are introduced, the system shall ensure that users can access only the information they are authorized to view or modify.

Acceptance criteria:

- Authentication is required where appropriate for protected resources.
- Authorization is enforced by the backend, not merely by hiding interface elements.
- Saved assessments are isolated according to the defined access policy.
- Access-control behavior is tested before deployment.

Authentication is not a mandatory MVP feature if the first version operates without accounts or persistent user-specific data.

AI Output Quality and Explainability

NFR-010: Structured Output Consistency

Priority: P0

AI-generated assessments shall follow a consistent structure so results can be displayed, validated, and evaluated reliably.

Acceptance criteria:

- Responses follow the defined CIRA-based schema.
- Impact categories and severity labels use the approved terminology.
- Required fields are checked before results are accepted.
- The application handles missing, malformed, or unexpected values safely.

NFR-011: Explainability

Priority: P0

The system shall make the reasoning behind potential impacts understandable to the intended user.

Acceptance criteria:

- Each relevant finding includes a rationale.
- Assumptions and information gaps are identified where possible.
- Users can distinguish the described impact from its possible consequences.
- The system does not present a plausible explanation as proof that a finding is factually

NFR-012: Uncertainty and Hallucination Management

Priority: P0

The system shall be designed to reduce unsupported claims and communicate uncertainty.

Acceptance criteria:

- The model is instructed to avoid inventing organizational facts.
- Findings that depend on unconfirmed details are marked for review.
- Missing information is surfaced where it materially affects the analysis.
- Human review remains necessary before findings are used for consequential decisions.

These controls reduce risk but cannot guarantee that hallucinations or incorrect assessments will never occur.

NFR-013: Repeatability and Evaluation

Priority: P0

The product team shall evaluate analysis quality using a consistent set of representative test cases.

Acceptance criteria:

- A test dataset includes different change types and levels of contextual detail.
- Expected findings or evaluation criteria are documented for each test case.
- Outputs are reviewed for relevance, coverage, unsupported claims, severity consistency, and actionability.
- Material changes to the model, prompt, or output schema trigger appropriate regression testing.

Because AI responses may vary between runs, identical inputs are not required to produce identical wording. They should, however, meet the defined quality and safety expectations.

Usability and Accessibility

NFR-014: Ease of Use

Priority: P0

The product shall be understandable to business analysts, project managers, product managers, and transformation professionals without requiring specialist AI knowledge.

Acceptance criteria:

- Labels and instructions use clear business language.
- Technical terminology is explained when necessary.
- Findings are organized into recognizable sections.
- Users can identify which findings require review and what actions are suggested.

NFR-015: Visual Accessibility

Priority: P1

The interface should support readable content and accessible interaction patterns.

Acceptance criteria:

- Text and background combinations provide sufficient contrast.
- Status and severity are communicated through text or labels, not color alone.
- Interactive elements have clear labels and visible focus states.
- Layouts remain usable on supported screen sizes.
- Keyboard navigation is supported for core interactions.

Accessibility should be tested against an appropriate standard, such as WCAG 2.2 Level AA, before claiming formal conformance.

Maintainability and Extensibility

NFR-016: Modular Architecture

Priority: P0

The application shall separate the user interface, application logic, AI integration, output validation, and optional data persistence into maintainable components.

Acceptance criteria:

- AI provider calls are handled through the server-side application layer.
- Output validation is separated from model generation.
- CIRA rules and permitted classifications are defined consistently.
- Changes to prompts or the model do not require unnecessary changes throughout the interface.

NFR-017: Version Control and Configuration

Priority: P0

The source code and relevant configuration shall be managed to support controlled development and maintenance.

Acceptance criteria:

- Source code is maintained in version control.
- Environment-specific settings are separated from application code.
- Prompt and schema changes can be tracked.
- Setup and deployment instructions are documented.

Scalability and Cost Management

NFR-018: Controlled Resource Usage

Priority: P1

The application should manage AI usage and computational resources in line with the expected scale of the MVP.

Acceptance criteria:

- Input and output sizes are bounded by defined limits.
- Unnecessary duplicate AI requests are minimized.
- AI usage and associated costs can be monitored where provider tooling permits.
- Rate limits and usage controls can be adjusted as demand changes.

NFR-019: Future Scalability

Priority: P2

- The architecture should allow the product to evolve without requiring a complete redesign for every new capability.

- Potential future extensions include document-based context retrieval, assessment history, enterprise authentication, and integration with project-management systems.
- These capabilities are architectural considerations, not claims that they are included in the initial release.

Non-Functional Acceptance Principle

- Non-functional requirements shall be verified through appropriate testing, measurement, configuration review, or documented inspection.
- Targets must be evaluated in the actual implementation environment. Security, privacy, reliability, and AI quality should not be assumed merely because the architecture describes them.
- Core principle: The product must not only generate an assessment; it must deliver that assessment in a way that is reliable, understandable, secure, and appropriate for human decision-making.

Evaluation Framework

Purpose

The Evaluation Framework defines how the AI Change Impact Analyzer will be tested and assessed before its results are considered useful for real-world decision support.

Since the product uses a Large Language Model (LLM), successful execution alone does not guarantee a high-quality assessment. The system must also identify relevant impacts, explain its reasoning, surface potential risks, acknowledge uncertainty, and produce actionable recommendations.

The framework evaluates the quality of AI-generated assessments while preserving human judgment as the final validation mechanism.

Evaluation Objectives

The evaluation process shall determine whether the system can:

- Identify relevant potential impacts across the six CIRA dimensions.
- Recognize meaningful risks and dependencies.
- Explain why a change may affect a particular business area.
- Distinguish supported findings from assumptions and information gaps.
- Assign severity and confidence labels consistently.
- Generate practical recommendations linked to identified findings.
- Produce structured output that meets the application's schema.
- Support effective human review without presenting AI-generated findings as confirmed facts.

Key Evaluation Dimensions

EV-001: Impact Coverage

Purpose: Evaluate whether the system identifies the important potential impacts associated with a proposed change.

Measurement approach:

Compare the generated assessment against a reference checklist of relevant impacts prepared or reviewed by a knowledgeable evaluator.

Success criteria:

- All six CIRA dimensions are considered.
- Important expected impacts are identified where supported by the supplied context.
- Missing impacts are recorded for review.
- The system does not invent findings simply to increase coverage.

EV-002: Finding Relevance

Purpose: Evaluate whether identified findings are meaningfully connected to the proposed change.

Measurement approach:
A reviewer assesses a sample of findings for relevance, duplication, and contextual support.

Success criteria:

- Findings have a clear relationship to the proposed change.
- Irrelevant or repetitive findings are minimized.
- Findings reflect the supplied context rather than generic business advice alone.
- Unsupported findings are identified during evaluation.

EV-003: Risk Identification Quality

Purpose: Evaluate whether the system identifies plausible risks arising from the change and its potential impacts.

Measurement approach:
Compare identified risks with a reference risk checklist and obtain reviewer feedback on relevance and completeness.

Success criteria:

- Important expected risks are identified where the available information supports them.
- Risks are distinguished from confirmed incidents or existing problems.
- Related impacts and dependencies are linked where appropriate.
- Unconfirmed risks are clearly presented as potential risks.

EV-004: Explainability and Traceability

Purpose: Determine whether users can understand why a finding was generated.

Measurement approach:
Review each sampled finding to assess the clarity of its rationale and its connection to the supplied change description or context.

Success criteria:

- Findings include an understandable rationale.
- Assumptions and information gaps are surfaced where relevant.
- Reviewers can trace findings back to the supplied information or a clearly identified inference.
- Explanations do not create a false impression of factual certainty.

EV-005: Severity and Confidence Consistency

Purpose: Evaluate whether severity and confidence classifications are applied consistently.

Measurement approach:
Compare classifications against documented CIRA definitions and have a reviewer assess a representative sample.

Success criteria:

- Severity labels follow the approved definitions.
- Confidence is assessed separately from severity.
- Classification rationales are available for review.
- Disagreements between the system and human reviewers are documented and analyzed.

EV-006: Recommendation Actionability

Purpose: Evaluate whether recommendations help users determine what to investigate, validate, mitigate, or monitor.

Measurement approach:
Review recommendations for specificity, relevance, feasibility, and connection to identified findings.

Success criteria:

- Recommendations describe a practical next step.
- Recommendations are linked to relevant impacts, risks, or information gaps.
- Suggested owners are included only when supported by context.
- Recommendations do not imply that an action has been approved or completed.

EV-007: Output Structure and Validation

Purpose: Confirm that AI responses can be processed safely and displayed consistently by the application.

Measurement approach:

Run generated responses through automated schema validation and application-level checks.

Success criteria:

- Valid responses conform to the defined output schema.
- Invalid or incomplete responses are handled safely.
- Severity, confidence, category, and review-status values follow approved formats.
- Structural errors do not result in misleading or falsely complete assessments.

EV-008: Uncertainty and Unsupported Claims

Purpose: Evaluate how well the system handles missing information and avoids presenting assumptions as facts.

Measurement approach:

Include test cases with incomplete context, ambiguous descriptions, and unknown dependencies.

Success criteria:

- Material information gaps are identified.
- Unsupported organizational details are not presented as confirmed facts.
- Findings requiring further investigation are flagged.
- The system communicates limitations without claiming that its assessment is exhaustive.

Evaluation Datase

The initial evaluation dataset shall contain representative change scenarios covering different business and technology contexts.

Example scenario categories include:

1. Process automation
2. Introduction of a new software platform
3. Modification of an existing business workflow
4. Integration between business systems
5. Changes to data collection or reporting
6. Organizational role or responsibility changes
7. Incomplete or ambiguous change descriptions

Each test case should include a proposed change, available context, expected evaluation considerations, and a reference checklist of important potential impacts and risks.

Reference checklists should be reviewed by a knowledgeable person where possible. They should be treated as evaluation aids rather than absolute proof that every possible finding has been identified.

Proposed Evaluation Metrics

The following metrics are proposed for the initial version. They are starting targets for testing, not claims of existing product performance.

Metric	Measurement	Initial target
Schema validity	Percentage of responses passing structural validation	100% of accepted responses
Impact relevance	Percentage of reviewed findings rated relevant	At least 85%
Reference impact coverage	Relevant expected impacts identified ÷ total expected impacts	At least 80%
Recommendation actionability	Percentage of reviewed recommendations rated actionable	At least 80%
Rationale adequacy	Percentage of reviewed findings with a clear, context-linked explanation	At least 85%
Unsupported-claim rate	Percentage of reviewed findings containing unsupported factual claims	As low as possible; target below 5%
Critical error rate	Percentage of assessments containing a materially misleading critical error	Target 0% in the evaluation set

These targets should be refined after establishing the test dataset, reviewer guidance, and baseline results. A zero-error target is an objective, not a guarantee that errors cannot occur in deployment.

For metrics based on human judgment, reviewers should use documented scoring criteria. Where feasible, more than one reviewer should assess a sample to identify differences in interpretation.

Evaluation Method

The evaluation process shall follow a repeatable sequence:

1. **Prepare test cases:** Create representative scenarios with different change types and levels of context.
2. **Define reference criteria:** Document expected impacts, risks, assumptions, and evaluation considerations.
3. **Run the assessment:** Submit each scenario through the application using the configured model and prompt.
4. **Validate the structure:** Check the response against the expected schema.
5. Review the findings: Evaluate relevance, coverage, rationale, severity, confidence, and recommendations.
6. **Record defects:** Document missing impacts, unsupported claims, inconsistent classifications, and technical failures.
7. **Improve the system:** Update prompts, validation rules, or application logic where justified.
8. **Repeat testing:** Re-run the relevant test cases after material changes and compare results with the baseline.

Human Evaluation and Feedback

Human evaluation is required because some aspects of impact analysis depend on business context, organizational knowledge, and professional judgment.

Reviewers should be able to classify findings as:

- Validated
- Not Relevant
- Needs Review

Evaluation feedback should also record why a finding was rejected, corrected, or retained where feasible.

Reviewer decisions can help identify recurring weaknesses in the system. If this feedback is used to change prompts, models, or evaluation rules, those changes should be documented and tested before release.

Human validation does not automatically guarantee correctness; reviewers must have sufficient context and appropriate expertise.

Regression Testing

Regression testing shall be performed whenever a material change is made to the AI prompt, model configuration, output schema, CIRA rules, or relevant application logic.

The regression process should verify that changes improve or preserve overall quality without introducing significant new errors.

Results should be compared against the established baseline, and important trade-offs should be documented.

Evaluation Reporting

Evaluation results should be summarized in a concise report containing:

- Test dataset description and size
- Model and prompt versions used
- Results for each evaluation metric
- Important failure cases and observed patterns
- Known limitations and unresolved issues
- Changes made in response to evaluation findings
- Recommendation on readiness for the next testing stage

The report must distinguish measured results from proposed targets and subjective reviewer observations.

Evaluation Reporting

Before a version is considered ready for its intended demonstration or pilot, the product team should confirm that:

- Core functional requirements have been tested.
- Accepted AI responses pass structural validation.
- Representative CIRA scenarios have been evaluated.
- Important unsupported-claim and classification errors have been reviewed.
- Security and data-handling controls have been checked.
- Known limitations are documented and visible where relevant.
- The product is clearly positioned as decision support, not an autonomous decision-maker.

A successful evaluation supports a release decision but does not prove that the system will identify every impact or risk in every real-world situation.

Core principle: The product should be evaluated on the quality and usefulness of its findings —not simply on whether the AI produces a complete-looking response.

Risks, Limitations & Mitigations

Purpose

The AI Change Impact Analyzer uses AI to identify potential impacts, risks, dependencies, and recommended actions associated with organizational change. However, AI-generated assessments may be incomplete, inaccurate, or based on assumptions.

This section identifies the key product, technical, operational, and AI-related risks that could affect the product's reliability, adoption, and usefulness.

The objective is not to eliminate every risk, which may not be possible, but to reduce foreseeable risks through appropriate design controls, testing, transparency, and human oversight.

AI and Analysis Risks

RISK-001: AI Hallucinations

Description: The AI may generate incorrect information, invent organizational details, or present assumptions as facts.

Potential impact: Users may make decisions based on misleading findings.

Likelihood: Medium to High
Impact: High

Mitigation:

- Instruct the model to distinguish supplied facts from assumptions and potential impacts.
- Require a rationale for each significant finding.
- Validate responses against the expected output schema.
- Flag uncertain findings for human review.
- Test the system using scenarios designed to reveal unsupported claims.

Residual risk: AI-generated findings may still be incorrect even after these controls are applied.

RISK-002: Missed Impacts or Dependencies

Description: The system may overlook important effects, especially when organizational context is incomplete or dependencies are complex.

Potential impact: Teams may underestimate the work, resources, or coordination required to implement a change.

Likelihood: Medium to High
Impact: High

Mitigation:

- Examine all six CIRA dimensions.
- Ask users to provide relevant organizational context.
- Identify missing information and unconfirmed dependencies.
- Use representative evaluation scenarios and reference checklists.
- Communicate that the assessment is not guaranteed to be exhaustive.

Residual risk: Some impacts may remain undiscovered without access to detailed organizational knowledge.

RISK-003: Inconsistent Severity Classification

Description: The AI may assign different severity levels to similar findings or classify severity without enough evidence.

Potential impact: Users may prioritize the wrong issues or overlook significant risks.

Likelihood: Medium
Impact: High

Mitigation:

- Define consistent severity criteria for Critical, High, Medium, and Low.
- Keep severity separate from confidence.
- Require a supporting rationale for classifications.
- Evaluate consistency across representative test cases.
- Allow users to review and challenge classifications.

Residual risk: Severity judgments may remain context-dependent and require professional assessment.

RISK-004: Generic or Low-Value Recommendations

Description: The AI may produce broad recommendations that sound reasonable but do not help users take a specific next step.

Potential impact: Users may spend additional time interpreting recommendations or lose confidence in the product.
Likelihood: Medium
Impact: Medium

Mitigation:

- Link recommendations to identified findings, risks, or information gaps.
- Prefer specific actions such as investigating, validating, updating, mitigating, or monitoring.
- Include responsible stakeholders only when supported by context.
- Evaluate recommendations for relevance, feasibility, and actionability.

Residual risk: Recommendations may require refinement to fit the organization's actual operating environment.

RISK-005: False Sense of Completeness

Description: A polished report with structured findings may appear comprehensive even when important information is missing.

Potential impact: Users may treat the assessment as a complete audit or implementation plan.
Likelihood: Medium
Impact: High
Mitigation:

- Display material limitations and information gaps.
- Identify findings that require review.
- Avoid claims that every impact or dependency has been identified.
- Distinguish AI-generated findings from human-validated findings.
- Reinforce the product's decision-support purpose.

Residual risk: Users may still overestimate the completeness of an assessment.

Data, Privacy, and Security Risks

RISK-006: Exposure of Sensitive Organizational Information

Description: Users may submit confidential business processes, internal system details, employee information, or other sensitive data to the application.

Potential impact: Unauthorized disclosure could create legal, contractual, reputational, or operational consequences.

Likelihood: Medium
Impact: High
Mitigation:

- Clearly explain how submitted information is processed.
- Minimize the collection and transmission of sensitive information.
- Evaluate the selected AI provider's data handling, retention, and training policies.
- Keep API credentials on the server side.
- Avoid logging confidential inputs or complete responses unnecessarily.
- Apply appropriate access controls and retention policies where data is stored.

Residual risk: Risk depends on the deployment environment, provider arrangements, and actual data-handling practices.

RISK-007: Unauthorized Access

Description: If the application introduces user accounts or saved assessments, weaknesses in authentication or authorization may expose protected information.

Potential impact: Users could access or modify assessments they are not authorized to use.

Likelihood: Medium
Impact: High
Mitigation:

- Enforce authorization on the backend.
- Apply access controls to stored assessments.
- Validate requests to protected resources.
- Test access restrictions before deployment.
- Avoid collecting user data that the MVP does not need.

Residual risk: Security vulnerabilities may still emerge and require ongoing testing and maintenance.

RISK-008: Prompt Injection and Malicious Input

Description: Submitted text or future document inputs may contain instructions intended to manipulate the AI or cause it to ignore the intended analysis rules.

Potential impact: The system may generate misleading results, misuse available context, or behave outside its intended boundaries.
Likelihood: Medium
Impact: Medium to High
Mitigation:

- Treat user-provided text and retrieved content as untrusted input.
- Separate system instructions from user-supplied context.
- Limit the AI's access to tools and external actions.
- Validate model outputs before displaying or processing them.
- Test adversarial inputs as part of security and evaluation testing.

Residual risk: Prompt-based controls alone cannot guarantee protection against every manipulation attempt.

Technical and Operational Risks

RISK-009: AI Provider or Network Failure

Description: The selected AI service may experience downtime, latency, rate limits, or connectivity problems.

Potential impact: Users may be unable to generate assessments when needed.
Likelihood: Medium
Impact: Medium to High
Mitigation:

- Implement appropriate timeouts and error handling.
- Display understandable failure messages.
- Support safe retries where appropriate.
- Monitor service errors and response times.
- Document external service dependencies.

Residual risk: The application may remain unavailable when a critical external dependency is unavailable.

RISK-010: Uncontrolled AI Usage Costs

Description: Repeated requests, long inputs, large outputs, or unexpected usage may increase operating costs.

Potential impact: The product may become expensive to maintain or difficult to scale within the intended budget.
Likelihood: Medium
Impact: Medium
Mitigation:

- Set reasonable input and output limits.
- Monitor model usage and cost where provider tooling permits.
- Avoid unnecessary duplicate requests.
- Use an appropriate model for the task and evaluate cost against quality.
- Establish usage limits before wider deployment.

Residual risk: Costs may change with provider pricing, model selection, and usage volume.

RISK-011: Invalid or Inconsistent AI Output

Description: The model may return malformed JSON, omit required fields, or use classifications that do not match the application's schema.

Potential impact: The interface may fail to display results correctly or may misinterpret findings.
Likelihood: Medium
Impact: Medium
Mitigation:

- Define a clear output schema.
- Validate responses on the server.
- Handle missing fields and invalid values explicitly.
- Use constrained output formats when supported by the chosen model.
- Test failure cases and recovery behavior.

Residual risk: Validation cannot establish that a structurally valid response is factually correct.

RISK-012: Scope Expansion

Description: Adding dashboards, authentication, document retrieval, integrations, historical analysis, and workflow management too early may increase complexity beyond the MVP's available time and resources.

Potential impact: Delivery may be delayed, core functionality may remain incomplete, or maintenance may become unnecessarily difficult.
Likelihood: High
Impact: Medium to High
Mitigation:

- Prioritize the core change-analysis workflow.
- Separate MVP requirements from future enhancements.
- Use acceptance criteria to control feature completion.
- Introduce additional capabilities only when a clear user need is established.
- Review scope changes against time, cost, and product value.

Residual risk: New technical or user requirements may still require scope adjustments.

Adoption and Human Decision-Making Risks

RISK-013: Overreliance on AI

Description: Users may accept AI-generated findings without reviewing their accuracy or relevance.

Potential impact: Incorrect recommendations or missed impacts may influence organizational decisions.
Likelihood: Medium
Impact: High
Mitigation:

- Default findings to Needs Review.
- Provide clear explanations and uncertainty indicators.
- Allow users to mark findings as Validated or Not Relevant.
- Communicate that the system supports rather than replaces professional judgment.
- Avoid automatically approving or implementing recommendations.

Residual risk: Human reviewers may still make mistakes or fail to challenge incorrect findings.

RISK-014: Low User Trust or Adoption

Description: Users may not trust the assessment if findings are generic, difficult to understand, or insufficiently connected to their context.

Potential impact: Users may abandon the product or disregard useful findings.
Likelihood: Medium
Impact: Medium
Mitigation:

- Use clear business language.
- Explain the rationale behind findings.
- Show relevant limitations and assumptions.
- Test the interface with representative users where possible.
- Use evaluation feedback to improve the quality and presentation of findings.

Residual risk: Adoption will depend on user needs, perceived value, organizational culture, and the quality of the implementation.

Risk Prioritization

The following table summarizes the initial risk priorities. Ratings are preliminary product-planning judgments and should be reviewed as implementation and testing provide more evidence.

Risk	Likelihood	Impact	Priority
AI hallucinations	Medium	High	Critical
Missed impacts or dependencies	Medium	High	Critical
Inconsistent severity classification	Medium	High	High
False sense of completeness	Medium	High	High
Sensitive data exposure	Medium	High	High
Unauthorized access	Medium	High	High, if persistent data is introduced
Prompt injection	Medium	Medium–High	High
Provider or network failure	Medium	Medium–High	High
Uncontrolled AI costs	Medium	Medium	Medium
Invalid AI output	Medium	Medium	Medium
Scope expansion	High	Medium–High	High
Overreliance on AI	Medium	High	High
Low user trust or adoption	Medium	Medium	Medium

Critical and High risks should receive explicit mitigation and testing attention before a pilot or broader release. The ratings should not be treated as measured probabilities.

Known Product Limitations

The initial version of the AI Change Impact Analyzer has the following anticipated limitations:

- Limited organizational context: The quality of findings depends partly on the information users provide.
- No guaranteed completeness: The system cannot guarantee that every relevant impact, risk, or dependency will be identified.
- Context-dependent judgments: Severity and prioritization may vary across organizations and situations.
- Model variability: Different runs or model versions may produce different findings.
- External service dependency: AI analysis depends on the availability and behavior of the selected provider.
- Limited organizational knowledge: Without connected documents or validated organizational data, the system cannot reliably know every internal process or dependency.
- Human review requirement: AI-generated assessments require review before being used to support consequential decisions.
- Implementation-dependent capabilities: Persistent history, authentication, document retrieval, integrations, and report export are not assumed to exist unless implemented and tested.

These limitations should inform product messaging, user expectations, testing, and future roadmap decisions.

Risk Monitoring and Ownership

Risks should be reviewed throughout development and testing rather than only at the end of the project.

For each material risk, the product team should record:

- Risk identifier and description
- Likelihood and impact
- Mitigation or control
- Test evidence or monitoring approach
- Current status
- Follow-up action, where required

For a solo-built MVP, the developer may manage these responsibilities directly. In a team setting, ownership can be assigned to the relevant product, engineering, security, or business stakeholder.

Risk Monitoring and Ownership

No mitigation should be considered effective solely because it has been documented. Controls must be implemented where required and tested against realistic failure scenarios.

Risks that remain unresolved should be documented and considered when deciding whether the product is suitable for a demonstration, pilot, or production environment.

Core principle: The analyzer should make change-related uncertainty easier to examine—not create a false impression that uncertainty has been eliminated.

Product Roadmap

Purpose

The Product Roadmap defines the planned evolution of the AI Change Impact Analyzer from an initial working product to a more capable and scalable decision-support solution.

The roadmap prioritizes core functionality, analysis quality, and user validation before introducing advanced capabilities.

Development will follow an iterative approach. Each phase should deliver a usable outcome, demonstrate measurable progress, and provide evidence to guide the next phase.

The roadmap represents proposed future work. Features should be considered complete only after they have been implemented and tested.

Roadmap Strategy

The roadmap follows four principles:

- Build the core first: Establish a working change-analysis workflow before adding advanced features.
- Validate before expanding: Evaluate the quality and usefulness of AI-generated findings before increasing complexity.
- Keep humans in control: Preserve human review and decision-making throughout product development.
- Use evidence to prioritize: Let user feedback, evaluation results, technical feasibility, and cost guide future investments.

Product Development Phases

Phase 1: Core MVP

Objective: Build a working version that accepts a proposed change and generates a structured impact assessment.

Priority: P0 — Must

Planned capabilities:

- Change description and context input.
- Server-side AI integration.
- CIRA-based analysis across all six impact dimensions.
- Structured impact, risk, dependency, and recommendation outputs.
- Severity and confidence classifications.

- Output-schema validation.
- Clear presentation of findings and limitations.
- Basic loading, validation, and error states.
- Human review statuses within the active assessment session.

Key deliverable: A functioning end-to-end analyzer that can be demonstrated using representative change scenarios.

Exit criteria:

- A user can submit a valid change description and receive an assessment.
- All six CIRA dimensions are considered.
- Accepted responses pass structural validation.
- Core error-handling behavior has been tested.
- AI-generated findings are clearly distinguished from human-validated findings.

Dependencies: Selected AI provider, application framework, server-side API implementation, and defined output schema.

Phase 2: Evaluation and Quality Improvement

Objective: Establish evidence that the analyzer produces relevant, understandable, and useful findings.

Priority: P0 — Required before broader validation

Planned capabilities:

- Representative test dataset.
- Reference checklists for expected impacts and risks.
- Evaluation of finding relevance and coverage.
- Testing of severity and confidence consistency.
- Assessment of recommendation actionability.
- Testing for unsupported claims and incomplete context.
- Regression testing for prompt and schema changes.
- Basic recording of evaluation results.

Key deliverable: An evaluation report documenting measured performance, important failure cases, and areas for improvement.

Exit criteria:

- Core evaluation scenarios have been executed.
- Proposed metrics from Section 18 have been measured where feasible.
- Material errors and recurring weaknesses have been documented.
- Important findings have informed improvements or explicit limitations.
- Results distinguish measured performance from proposed targets.

Dependencies: A working MVP, documented evaluation criteria, and reviewer guidance.

Phase 3: Usability Testing and Pilot Validation

Objective: Determine whether the analyzer is useful to its intended users in realistic business scenarios.

Priority: P1 — Should Have

Planned capabilities:

- Usability testing with representative users where available.
- Feedback collection on clarity, relevance, and trust.
- Review of the change-input experience.
- Assessment of whether users can understand and validate findings.
- Refinement of explanations, navigation, and information hierarchy.
- Identification of the highest-value improvements.

Key deliverable: A pilot findings report containing user feedback, usability observations, and prioritized product improvements.

Exit criteria:

- Representative users or reviewers have completed relevant tasks where access permits.
- Confusing or misleading interactions have been documented.
- Feedback has been translated into prioritized changes.
- The product's limitations are communicated clearly.

Dependencies: A stable MVP and access to suitable testers. If external users are unavailable, structured internal testing can be used, with its limitations documented.

Phase 4: Product Refinement and Persistence

Objective: Determine whether the analyzer is useful to its intended users in realistic business scenarios.

Priority: P1 — Should Have

Planned capabilities:

- Usability testing with representative users where available.
- Feedback collection on clarity, relevance, and trust.
- Review of the change-input experience.
- Assessment of whether users can understand and validate findings.
- Refinement of explanations, navigation, and information hierarchy.
- Identification of the highest-value improvements.

Key deliverable: A pilot findings report containing user feedback, usability observations, and prioritized product improvements.

Exit criteria:

- Representative users or reviewers have completed relevant tasks where access permits.
- Confusing or misleading interactions have been documented.
- Feedback has been translated into prioritized changes.
- The product's limitations are communicated clearly.

Dependencies: A stable MVP and access to suitable testers. If external users are unavailable, structured internal testing can be used, with its limitations documented.

Phase 5: Advanced Context and Integrations

Objective: Improve the relevance of assessments by introducing richer organizational context and selected integrations.

Priority: P2 — Future enhancement

Potential capabilities:

- Document-based context retrieval using Retrieval-Augmented Generation (RAG).
- Analysis grounded in approved organizational documents.
- Integration with selected project-management or business systems.
- More detailed dependency mapping.

- Improved assessment comparison and traceability.
- Additional enterprise access and governance controls.

Key deliverable: A context-aware analyzer that can use approved organizational information beyond the user's manually entered description.

Exit criteria:

- A validated use case justifies the additional complexity.
- Data permissions and source traceability are defined.
- Retrieved information can be distinguished from AI-generated inferences.
- The system is evaluated for relevance, source accuracy, and privacy.
- Integration failures and access restrictions are handled safely.

Dependencies: User demand, suitable data sources, integration access, and additional engineering capacity.

Roadmap Summary

Phase	Focus	Primary outcome	Priority
1	Core MVP	Working CIRA-based analyzer	P0
2	Evaluation	Measured quality and documented limitations	P0
3	Pilot validation	User feedback and usability improvements	P1
4	Persistence and refinement	Repeat-use capabilities, if justified	P1/P2
5	Advanced context and integrations	Richer organizational intelligence	P2

The phases are ordered by dependency, not fixed calendar dates. Some testing and refinement activities may overlap.

Proposed Delivery Approach

For an initial solo-built implementation, the roadmap should be managed through short development iterations.

Each iteration should include:

1. Select a small, clearly defined outcome.
2. Implement the relevant functionality.
3. Test the behavior against its acceptance criteria.
4. Record defects and limitations.
5. Update the product backlog based on findings.
6. Demonstrate the working result before expanding scope.

Calendar estimates should be added only after the implementation environment, available time, and MVP boundaries have been confirmed.

Prioritization Rules

Future features should be evaluated against the following factors:

- **User value:** Does the feature solve a demonstrated problem?
- **Analysis quality:** Does it improve the relevance, completeness, or explainability of findings?
- **Risk reduction:** Does it improve privacy, reliability, or decision quality?
- **Implementation effort:** Can it be delivered within available time and resources?
- **Technical dependencies:** Does it require infrastructure or data that is not yet available?
- **Evaluation readiness:** Can its value and potential failure modes be tested?

Features that add complexity without demonstrating meaningful user value should be deferred.

Roadmap Success Measures

Roadmap progress should be assessed through observable deliverables and evidence rather than feature counts alone.

Key measures include:

- Completion of the end-to-end MVP workflow.
- Successful validation of accepted AI responses.
- Evaluation results against the Section 18 metrics.
- Number and severity of unresolved defects.
- Findings from usability testing.
- User-reported usefulness of the assessment.
- Demonstrated value of any feature introduced in later phases.

All numerical targets should be defined and evaluated using the agreed test approach. Planned features and target outcomes must not be represented as completed achievements.

Roadmap Governance

The roadmap should be reviewed whenever evaluation results, user feedback, technical constraints, or security considerations materially change the product's priorities.

Changes to the roadmap should document:

- What has changed.
- Why the change is necessary.
- The expected user or product benefit.
- The impact on scope, dependencies, and effort.
- How the change will be validated.

For the initial release, the recommended priority is to complete and evaluate the core analyzer before investing in persistent history, document retrieval, or enterprise integrations.

Core principle: Build a reliable and useful decision-support workflow first; expand only when evidence justifies the next investment.

Go-to-Market Strategy

Purpose

The Go-to-Market (GTM) Strategy defines how the AI Change Impact Analyzer will be introduced to its initial target users, how its value will be communicated, and how early feedback will guide future development.

The initial strategy focuses on demonstrating product value and validating user needs before pursuing broader adoption or commercialization.

This is a proposed GTM plan, not a claim of an existing customer base, market traction, or commercial launch.

Target Audience

The initial audience consists of professionals who assess business changes, coordinate implementations, or evaluate operational risks.

Target Segment	Primary Need
Business Analysts	Identify downstream impacts, dependencies, and requirement changes.
Project Managers	Understand implementation risks and cross-team dependencies.
Product Managers	Assess the wider consequences of product and process changes.
Digital Transformation Teams	Evaluate operational readiness and organizational implications.
Process and Operations Teams	Identify affected workflows, documentation, and responsibilities.

Initial focus: Business Analysts and project or product professionals involved in process improvement, software implementation, and digital transformation.

These users are a practical starting point because their work often requires understanding how a change in one area affects other parts of an organization

Positioning and Value Proposition

Product Positioning

The AI Change Impact Analyzer is an AI-assisted decision-support tool that helps professionals identify, organize, and evaluate the potential ripple effects of organizational change.

Core Value Proposition

Understand the wider impact of a change before implementation begins.
The product aims to help users:

- Identify potential impacts across six business dimensions.
- Surface risks and dependencies that may otherwise be overlooked.
- Understand the reasoning behind generated findings.
- Turn analysis into practical validation and mitigation actions.
- Review AI-generated findings before using them in business decisions.

Differentiation Hypotheses

The product's potential differentiation lies in its structured approach to change analysis rather than simply generating a general AI response.

Its proposed differentiators are:

- **CIRA framework:** A consistent structure for Change, Impact, Risk, and Action.
- **Six-dimensional analysis:** Processes, People, Systems, Data, Documentation, and Dependencies.
- **Explainable findings:** Each finding should include a rationale and potential consequences.
- **Human-led validation:** Users retain responsibility for confirming findings and making decisions.
- **Action-oriented output:** Findings are connected to suggested investigation, validation, mitigation, or monitoring activities.

These are intended product differentiators. Their effectiveness must be tested against user needs and alternative approaches.

Early User Acquisition Strategy

The initial acquisition strategy prioritizes low-cost channels that support direct feedback and product demonstration.

Channel	Approach	Objective
Portfolio Website	Publish a product overview, demonstration, and example assessment.	Establish credibility and communicate the product's purpose.
LinkedIn	Share change-analysis examples, product decisions, and lessons from evaluation.	Reach relevant professionals and invite discussion.
Professional Communities	Participate in Business Analysis, Product Management, and digital transformation communities.	Discover user problems and identify potential testers.
Direct Outreach	Contact relevant professionals with a concise product demonstration request.	Recruit early reviewers and gather specific feedback.
Pilot Demonstrations	Walk users through realistic change scenarios.	Observe how users interpret and apply the findings.

The initial approach should rely on existing professional networks, a portfolio demonstration, and direct conversations rather than paid advertising.

Outreach should communicate the product's current capabilities accurately and avoid presenting planned features as available functionality.

Pilot Validation Plan

Before pursuing a wider launch, the product should be evaluated through a small, structured pilot.

Proposed Pilot Activities

1. Select representative change scenarios, such as a new software implementation, workflow redesign, or system integration.
2. Ask participants to submit a change description and review the generated assessment.
3. Observe whether they can understand the findings, identify relevant risks, and validate the recommendations.
4. Collect feedback on relevance, clarity, trust, and practical usefulness.
5. Compare feedback and observed issues against the evaluation framework in Section 18.
6. Prioritize improvements before expanding the audience.

Pilot Success Criteria

The pilot should establish whether:

- Users understand the product's purpose and limitations.
- Findings are relevant to the scenarios being assessed.
- Explanations support meaningful human review.
- Recommended actions are practical and understandable.
- Users can identify missing context or incorrect findings.
- Feedback reveals a clear reason to use the product again.

The pilot size and numerical success thresholds should be set before testing. Results must reflect actual observations rather than assumed outcomes.

If external participants are unavailable, structured scenario testing can provide an initial evidence base, but it should not be described as validated customer adoption.

Adoption and Performance Metrics

The GTM strategy will use a small set of measures to evaluate awareness, product usefulness, and early adoption.

Metric	What It Measures
Demo Engagement	Whether intended users view or complete the product demonstration.
Assessment Completion Rate	The proportion of started assessments that produce a completed result.
Finding Review Rate	The proportion of generated findings that users review or classify.
Perceived Usefulness	User feedback on whether the assessment supports their work.
Finding Relevance	The proportion of reviewed findings considered relevant to the scenario.
Repeat Usage	Whether users return to assess additional changes, where usage can be tracked appropriately.
Feedback-to-Improvement Rate	The extent to which collected feedback leads to implemented improvements.

Not every metric requires product analytics. During an early pilot, structured feedback forms and manual observations may be sufficient.

Any reported metric must include its measurement method, sample size, and relevant limitations. Metrics that have not been measured should remain proposed measures.

Commercialization Hypotheses

Commercialization should be considered only after the product demonstrates meaningful value and a clearer understanding of its target users.
Potential models for future exploration include:

- **Free individual access:** A way for professionals to explore the core workflow.
- **Paid professional plan:** Potential access to advanced assessment management, exports, or repeat-use features.
- **Enterprise licensing:** Potential deployment with organizational context, access controls, governance, and integrations.

These are commercial hypotheses, not confirmed pricing or committed product tiers.

Before selecting a model, the product should investigate willingness to pay, frequency of use, organizational purchasing needs, competing alternatives, and the cost of operating the service.

Enterprise capabilities should not be marketed as available unless they have been implemented and validated.

GTM Risks and Mitigations

Risk	Mitigation
Users perceive the tool as a generic AI wrapper.	Demonstrate the CIRA framework, structured findings, and review workflow.
Early users distrust AI-generated findings.	Explain reasoning, communicate uncertainty, and preserve human validation.
The product promises more than it delivers.	Clearly separate available capabilities from the roadmap.
Initial feedback is too limited to represent the market.	Document the sample, test different scenarios, and avoid broad conclusions.
Users find the workflow valuable but do not return.	Investigate frequency of need and improve the workflow based on evidence.
Commercial demand remains unproven.	Test willingness to pay before investing in paid tiers or enterprise features.

Go-to-Market Success Definition

The initial GTM effort will be considered informative if it establishes:

- A clearly defined initial audience.
- A compelling and understandable product value proposition.
- A working demonstration of the core workflow.
- Evidence about relevance, usability, and user trust.
- A prioritized list of product improvements based on feedback.
- An initial view of repeat-use potential and commercialization opportunities.

Core principle: Demonstrate value, learn from real users, and expand only when evidence supports the next step.

Example Analysis

Purpose

This section demonstrates how the AI Change Impact Analyzer could evaluate a proposed business change using the Change Impact & Risk Analysis (CIRA) framework.

The example illustrates the expected workflow, output structure, reasoning, risk classification, and human-review process.

Important: This is an illustrative example, not a record of an actual production run or measured product performance. The findings are hypothetical and would require validation in a real organizational context.

Example Scenario

Proposed Change: Implement a New Employee Leave Management System

An organization currently manages employee leave requests through email and spreadsheets. It proposes introducing a centralized leave management system that allows employees to submit requests digitally, managers to approve or reject requests, and HR to maintain leave records.

The new system may integrate with the organization's HR database and payroll process.

Input Context

Field	Example Input
Change Type	Software implementation
Business Objective	Reduce manual work and improve leave-request visibility.
Current State	Requests are handled through email and spreadsheets.
Proposed State	Requests are submitted and tracked in a centralized system.
Affected Teams	HR, employees, line managers, payroll, and IT.
Known Dependencies	Employee records, approval rules, leave balances, and payroll processes.
Constraints	Existing policies and payroll procedures must remain accurate.
Missing Information	Integration details, migration approach, access permissions, and approval exceptions.

The missing information should remain visible because it may materially affect the assessment.

CIRA Analysis

C — Change

The proposed change replaces a manual leave-request process with a centralized digital workflow. The analysis should identify what is changing, why the change is proposed, which activities are affected, and what information is still unknown.

Initial interpretation: The change affects more than the user interface. It may also alter approval responsibilities, employee data handling, leave-balance calculations, and payroll inputs.

I — Impact

The analyzer should examine all six impact dimensions.

Dimension	Potential Impact	Why It Matters
Processes	Leave submission, approval, escalation, and recordkeeping may need redesign.	Existing procedures may no longer match the digital workflow.
People	Employees, managers, and HR staff may need training or new responsibilities.	Adoption depends on users understanding the new process.
Systems	The new platform may need to connect with HR and payroll systems.	Integration failures could disrupt downstream operations.
Data	Employee records, leave balances, and historical requests may need migration or synchronization.	Incorrect or incomplete data could produce inaccurate leave balances.
Documentation	Leave policies, SOPs, user instructions, and training materials may require updates.	Outdated documentation could lead to inconsistent practices.
Dependencies	Approval rules, employee master data, payroll cutoffs, and access permissions may affect implementation.	Unresolved dependencies could delay deployment or create operational errors.

These are potential impacts inferred from the scenario. Their relevance must be confirmed against the organization's actual processes and systems.

R — Risk

The following risks illustrate how the analyzer could prioritize areas requiring further investigation.

Risk ID	Risk	Severity	Confidence	Rationale
R-01	Incorrect leave balances after data migration or synchronization.	High	Medium	Leave balances depend on accurate historical and current records; the migration approach is unknown.
R-02	Payroll discrepancies caused by incorrect leave information or integration failures.	High	Medium	Payroll may rely on leave data, but the exact integration and reconciliation process has not been confirmed.
R-03	Approval delays due to unclear responsibilities or exception handling.	Medium	Medium	The digital workflow may change how requests are routed and escalated.
R-04	Unauthorized access to employee leave information.	High	Medium	Access permissions and role-based controls have not been specified.
R-05	Low adoption because users are unfamiliar with the new process.	Medium	Medium	Employees, managers, and HR may need communication and training.

Severity and confidence are different. Severity describes the potential consequence if a risk occurs. Confidence describes how strongly the available information supports the finding. Neither classification should be treated as a statistically calibrated probability unless the product has been specifically validated for that purpose.

The severity ratings above are provisional and should be adjusted to the organization's risk criteria.

A — Action

Each finding should lead to a practical next step.

Action ID	Recommended Action	Purpose
A-01	Reconcile migrated leave balances against the existing HR records before launch.	Reduce the risk of incorrect balances.
A-02	Test payroll-related scenarios and establish a reconciliation procedure.	Identify discrepancies before they affect payroll.
A-03	Document approval rules, escalation paths, and exception handling.	Clarify responsibilities and reduce delays.
A-04	Define role-based access permissions and test unauthorized-access scenarios.	Protect employee information.
A-05	Prepare role-specific training and provide a support route for launch.	Support adoption and issue resolution.
A-06	Confirm system dependencies, data ownership, and integration responsibilities with the relevant teams.	Resolve unknowns before implementation.

These actions are recommendations for investigation and planning. They do not imply that the actions have been approved, assigned, or completed.

Example Structured Finding

The following JSON illustrates how a single finding could be represented in a machine-readable output.

```json
{
  "finding_id": "IMP-004",
  "category": "Systems",
  "affected_area": "Payroll integration",
  "description": "The new leave management system may affect payroll inputs if leave data is transferred incorrectly or becomes inconsistent.",
  "severity": "High",
  "confidence": "Medium",
  "rationale": "The scenario identifies a potential relationship between leave records and payroll, but the integration design is not yet confirmed.",
  "potential_consequences": [
    "Incorrect payroll calculations",
    "Additional reconciliation work",
    "Delayed resolution of employee pay queries"
  ],
  "related_risks": [
    "R-02"
  ],
  "recommended_action": "Validate payroll integration scenarios and define reconciliation controls before launch.",
  "validation_status": "Needs Review"
}
```

The structure is illustrative and should be aligned with the actual schema defined in Section 15 and the functional requirements in Section 16.

Human Review Workflow

AI-generated findings should enter the review process with the default status Needs Review.

Review Status	Meaning	Expected Behavior
Needs Review	The finding has been generated but not yet assessed by a reviewer.	Review the rationale, context, and potential consequences.
Validated	A reviewer has confirmed that the finding is relevant to the organizational context.	Use it as an input to planning, subject to normal governance and approval processes.
Not Relevant	A reviewer has determined that the finding does not apply to the current scenario.	Record the decision where supported and exclude it from the applicable impact assessment.

Reviewing a finding does not automatically approve a project, authorize a change, or execute a recommended action.

If the product does not yet support persistent review records, these statuses should be treated as session-level states rather than permanent audit records.

Example Output Summary

A completed assessment for this scenario could present the following summary:

- **Change:** Introduce a centralized employee leave management system.
- **Impact coverage:** Processes, People, Systems, Data, Documentation, and Dependencies.
- **Priority investigation areas:** Leave-data accuracy, payroll integration, access permissions, and approval workflows.
- **Key unknowns:** Integration design, migration controls, exception handling, and access configuration.
- **Immediate next step:** Validate critical data and system dependencies with HR, IT, payroll, and process owners.
- **Decision authority:** The organization's responsible stakeholders, not the AI analyzer.

This summary should link back to the detailed findings so users can understand the basis of each conclusion

What This Example Demonstrates

The example illustrates the intended product behavior:

1. Convert a business change description into a structured assessment.
2. Examine the change across all six impact dimensions.
3. Identify potential risks and dependencies.
4. Explain why each finding may matter.
5. Distinguish severity from confidence.
6. Recommend actions that users can investigate or validate.
7. Preserve human review and decision-making.
8. Make missing information and uncertainty visible.

The example does not establish that the analyzer will identify every impact or risk. Actual performance must be assessed using the evaluation framework in Section 18.

Core principle: The analyzer should help teams ask better questions, investigate overlooked dependencies, and make more informed change decisions—not present AI-generated findings as unquestionable facts.

Glossary & References

Purpose

This section defines the key terms, abbreviations, frameworks, and technical concepts used throughout the AI Change Impact Analyzer Product Requirements Document (PRD).
It also identifies the reference materials that inform the product's design, implementation, evaluation, and responsible use of AI.

The glossary supports consistent interpretation across product, business, technical, and evaluation discussions.

Glossary

Term	Definition
AI (Artificial Intelligence)	Technology that performs tasks such as generating content, identifying patterns, and supporting analysis.
AI-Assisted Decision Support	The use of AI to help people analyze information and consider possible decisions while retaining human authority.
API (Application Programming Interface)	A defined interface that allows software systems to communicate with one another.
Business Analyst (BA)	A professional who investigates business needs, processes, requirements, and potential solutions.
CIRA	Change Impact & Risk Analysis, the framework used by the analyzer to structure change analysis into Change, Impact, Risk, and Action.
Change Impact Analysis	The process of identifying how a proposed change may affect business processes, people, systems, data, documentation, and dependencies.
Confidence	An indicator of how strongly the available context supports a finding. It is not necessarily a calibrated probability.

Dependency	A relationship in which an activity, process, system, or outcome relies on another component or condition.
Functional Requirement	A statement describing what the product must do.
Hallucination	AI-generated content that is inaccurate, unsupported, or presented as factual without adequate evidence.
Human-in-the-Loop	An approach in which people review, validate, or guide AI-generated outputs and retain responsibility for relevant decisions.
Impact Finding	A structured record describing a potential consequence of a proposed change, including its rationale and recommended action.
Large Language Model (LLM)	An AI model trained to process and generate language-based content.
Non-Functional Requirement	A statement describing how the product should perform or the quality, security, usability, or operational constraints it must satisfy.
Prompt Injection	An attempt to manipulate an AI system through instructions embedded in user input or other content it processes.
RAG (Retrieval-Augmented Generation)	An approach that retrieves relevant information from external or organizational sources to provide additional context to an AI model.
Risk	A potential event or condition that could cause an undesirable consequence.
Severity	A classification of the potential consequence or impact of a risk.
Structured Output	AI-generated information returned in a defined format, such as a JSON object that follows a specified schema.

Validation	The process of checking whether a finding, output, or requirement is correct, relevant, and suitable for its intended use.
Zod	A TypeScript-oriented schema validation library that can be used to validate application data.

References

The following resources provide the conceptual and methodological foundation for the AI Change Impact Analyzer. They inform the product's approach to business analysis, requirements engineering, organizational change, risk management, and responsible AI.

Business Analysis and Requirements Engineering

1. **International Institute of Business Analysis (IIBA)** - Business analysis practices, requirements analysis, stakeholder engagement, and solution evaluation.

→ Link: https://www.iiba.org/

2. **IIBA — A Guide to the Business Analysis Body of Knowledge (BABOK® Guide) -** Reference for business analysis practices, including requirements lifecycle management, strategy analysis, and solution evaluation.

→ Link: https://www.iiba.org/standards-and-resources/babok/

3. **International Organization for Standardization (ISO) — ISO/IEC/IEEE 29148:2018 -** Guidance on requirements engineering, including requirements processes and requirements specification.

→ Link: https://www.iso.org/standard/72089.html

Change Management and Organizational Change

4. **Prosci — Change Management Resources -** Practical guidance on organizational change, stakeholder readiness, adoption, and the people side of change.

→ Link: https://www.prosci.com/resources

5. **Association of Change Management Professionals (ACMP)** - Professional resources and standards relating to structured change management practices.

→ Link: https://www.acmpglobal.org/

Risk Management and Impact Assessment

6. **International Organization for Standardization (ISO) — ISO 31000:2018, Risk Management -** Principles and guidelines for identifying, assessing, treating, and monitoring risk.

→ Link: https://www.iso.org/iso-31000-risk-management.html

7. **P**roject Management Institute (PMI) - Resources on project risk, stakeholder management, project delivery, and managing change within initiatives.

→ Link: https://www.pmi.org/

Artificial Intelligence and Responsible AI

8. **National Institute of Standards and Technology (NIST) — AI Risk Management Framework -** Guidance for identifying, assessing, and managing risks associated with AI systems.

→ Link: https://www.nist.gov/itl/ai-risk-management-framework

9. **OWASP — Top 10 for Large Language Model Applications -** Security guidance for AI applications, including prompt injection, insecure output handling, and other LLM-related risks.

→ Link: https://owasp.org/www-project-top-10-for-large-language-model-applications/

10. **Google — People + AI Guidebook -** Practical guidance for designing human-centered AI experiences, communicating AI capabilities, and supporting appropriate user trust.

→ Link: https://pair.withgoogle.com/guidebook/

Reference Usage Note

These resources inform the product's design principles and analytical approach. They do not establish that the CIRA framework is an externally standardized framework or that the analyzer is certified, compliant, or independently validated against the referenced standards.

References should be checked for current editions and accessibility before the PRD is formally published. Where a specific standard or publication directly informs a requirement, the relevant edition or section should be recorded where practical.

END OF DOCUMENT

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